CivicConnect

Project Engineering Document v4.0 - review candidate

Engineering foundation, architecture, construction and verification

Full continuing PED - 2026-10-18
SEN381 Software Engineering 381
Belgium Campus ITversity

Team: Group U - Emile, Don and Masego

M3 implementation baseline: c7e92de8f818b66ca1ecc0faf88b0987c4b17b40
Merged through PR #85. The recorded approvals concern that implementation slice. Check the associated documentation PR for this export's review status.

Repository: github.com/DonLukus/Civic-connect

Continuity: This export includes every numbered Markdown source currently present in docs/PED, starting with document control, preserving M1/M2/M3, and continuing through section 23. Earlier statements and signatures retain their dated scope. Read the latest dated closure section for current status.

Source boundary: The repository has no section 18 source. Existing numbering is preserved; this export does not invent the missing technology narrative or promote any baseline/ADR status.

Assessment access: RTM: docs/requirements/RTM.csv. CI/test evidence and local staging instructions are indexed in section 22. Public production readiness: NOT READY. No new signatures appear in this export.

Submission boundary: Deadline 18 October 2026, supplied by Emile. Final team approval, deployment decision, assessed documentation reference and actual Moodle submission remain pending. This review export is not a signed Final submission; section 23 contains the proposed original-AC scope result and unresolved obligations.

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Document Control

Source: docs/PED/00-document-control.md

PED section, v1.0 baselined, v2.0 not yet baselined, now also opening v3.0. Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` at M1, `PED_v2.0.docx` at M2, `PED_v3.0.docx` at M3 (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

SEN381 Software Engineering 381 - Project Engineering Document

CivicConnect - Community Service Request Management Platform Version 3.0 (in progress) - Construction & Verification · Milestone 3, continuing from the v2.0 Milestone 2 work below, which is itself **not yet BASELINED** (see the graduation checklist further down this document - recorded honestly, not glossed over)

Field	Record
Module	SEN381 Software Engineering 381 (NQF 8), Belgium Campus ITversity
Team (M1 workstreams)	Masego (Workstream A) · Don (Workstream B) · Emile (Workstream C)
Team (M2 members)	Emile (Member A) · Don (Member B) · Masego (Member C) - see note below, the lettering changes between milestones
Team (M3 roles)	Don - Backend & configuration · Masego - Verification & traceability · Emile - UI, staging, performance & release evidence
Repository	https://github.com/DonLukus/Civic-connect
Baseline date	M1: 8 September 2026 · M2: in progress · M3: in progress
Governing document	SEN381 CivicConnect Master Project Brief v1.1 (2026); SEN381 CivicConnect Milestone 2 brief; SEN381 CivicConnect Milestone 3 brief
Status	M1 baselined and gated. M2 underway: see the v2.0 graduation checklist below for exactly what's still open. M3 construction has started in parallel (agreed M3 journey scope: login, submit request, view own requests/status, staff queue and accept) - starting construction before M2's baseline fully closes is a real, deliberate sequencing choice, not an oversight, and is recorded here rather than hidden.

Document Control

0.1	
Ver	0.1
Date	2026-09-08
Author	Masego
Change	Problem, stakeholders, scope, constraints, decisions
Reviewed by	Don, Emile
Status	Draft

0.2	
Ver	0.2
Date	2026-09-08
Author	Don
Change	Requirements, acceptance criteria, RTM, forward engineering, repository governance
Reviewed by	Masego, Emile
Status	Draft

0.3	
Ver	0.3
Date	2026-09-08
Author	Emile
Change	Risk, AI control, working agreement, document control
Reviewed by	Masego, Don
Status	Draft

0.9	
Ver	0.9
Date	2026-09-08
Author	Emile
Change	Sections integrated; identifiers reconciled across all registers
Reviewed by	Masego, Don
Status	In review

1.0	
Ver	1.0
Date	8 September 2026
Author	Team
Change	Baselined. Scope, requirements and acceptance criteria under change control.
Reviewed by	All three
Status	BASELINED

2.0 (in progress)	
Ver	2.0 (in progress)
Date	2026-09-30
Author	Emile
Change	M2 document control opened. PED continues from the v1.0 baseline - nothing below is rewritten, M2 adds new sections (16-21) for architecture, data, technology, design and the M2 sign-off.
Reviewed by	Don, Masego (pending)
Status	Draft

2.0 review candidate	
Ver	2.0 review candidate
Date	2026-10-01
Author	Emile
Change	Reconciled PoC, campus test, PR #62 and Render deployment evidence with the architecture and deployment sections.
Reviewed by	Don, Masego (pending)

2.0 review candidate	
Status	In review

3.0 (in progress)	
Ver	3.0 (in progress)
Date	2026-10-15
Author	Emile
Change	M3 document control opened. PED continues from v2.0 - nothing above is rewritten. New sections numbered from 22 onward will cover construction, verification and release evidence as they land; none exist yet, this row only records that the version is open.
Reviewed by	Don, Masego (pending)
Status	Draft

Note on the M2 team table above: the Member A/B/C split for Milestone 2 is not the same grouping as the Workstream A/B/C split above it - those were fixed at M1 baseline and stay as the historical record. M2 uses its own lettering, matching the assessment brief's grading criteria (Emile: criteria A + B, architecture and requirements; Don: criteria C + E, data and design patterns; Masego: criteria D + F, technology and delivery).

Authorship, review and approval

Member	Authored	Reviewed	Approval
Masego	§2, 3, 4, 5, 10	§6, 7, 8, 9, 11-13	Approved 8 September 2026
Don	§6, 7, 9, 12	§2-5, 10, 11, 13	Approved 8 September 2026
Emile	§1, 8, 11, 13, 14, 15	§2-7, 9, 10, 12	Approved 8 September 2026

Every section was reviewed by two members other than its author, through pull requests. Authorship identifies who drafted a section; all three members are accountable for the whole document. The gate decision rests with the assessor.

Linked controlled artefacts

Registers reproduced in this document are extracts taken at baseline. The CSV set in the repository is the source of truth (DEC-005). The "State at M2 (in progress)" column is added here, not substituted for the v1.0 column, per the same never-overwrite rule that governs the rest of this document.

Artefact	Location	State at v1.0	State at M2 (in progress)
Requirements, scope, constraints	docs/requirements/	Baselined	Unchanged except CHG-001 (§10.3)
Traceability matrix	docs/requirements/RTM.csv	40 of 40 traced	M2 columns added; 3 of 40 rows carry real M2 evidence, the rest read Pending M2 - tracked on Issue #66
Risk and assumption registers	docs/risk/	14 risks, 5 assumptions	16 risks (RSK-15, RSK-16 added)
Decision log	docs/decisions/	10 decided, 2 deferred	**Gap:** 6 new ADRs exist in docs/decisions/adr/ (ADR-001, 002, 003, 004, 006, 007) but none has a corresponding DEC- row yet in docs/decisions/decision-log.csv or this document's §10 table - tracked on Issue #65
ADR set	docs/decisions/adr/	Did not exist	6 ADRs: Accepted - ADR-002, 003, 004, 006; Proposed - ADR-001, ADR-007

Artefact	Location	State at v1.0	State at M2 (in progress)
Forward engineering register	docs/forward-engineering/	7 considerations	Unchanged
AI usage register, governance, sign-off	docs/governance/, docs/baseline/	Live and controlled	M2 baseline sign-off recorded in docs/baseline/M2-baseline-signoff.md - see §21 and the graduation checklist below

When v2.0 moves from Draft to BASELINED

This document does not assert v2.0 is baselined - it stays Draft/In review until every item below is objectively true, verified against the actual controlled artefact named, not asserted here. This mirrors exactly how v1.0 graduated: a final `BASELINED` row gets appended below the current "2.0 review candidate" row (never edited in place - the draft history stays visible), once:

#	Condition	Verified against
1	ADR-001 status is Accepted, not Proposed	`docs/decisions/adr/ADR-001-architecture-and-stack-options.md`
2	ADR-007 status is Accepted, not Proposed	`docs/decisions/adr/ADR-007-authentication-authorization-placement.md`
3	DEC-008's evidence (login/save/deploy on a live host, verified not assumed) exists	`docs/decisions/poc-log.md`
4	Every M2 ADR has a corresponding DEC- row	`docs/decisions/decision-log.csv` + §10
5	§18 (technology stack narrative) exists	`docs/PED/18-technology-stack.md` - outstanding, Masego (Issue #65/#66 overlap)
6	RTM M2 columns are populated for all 40 rows, not just the traced slice (implementation evidence may still legitimately read Pending for unbuilt requirements - architecture/data/design/technology links may not)	`docs/requirements/RTM.csv` - Issue #66
7	`docs/baseline/M2-baseline-signoff.md`'s own Outcome field reads ACCEPTED, signed by all three	that file directly

Once all seven hold, the final Document Control row and this section both get a follow-up PR updating "Status" (row above) and the Baseline date field to the real M2 date - not before.

M3 review update - 16 October 2026

Section 22 and `docs/release/m3-release-evidence-summary.md` index the integrated local verification candidate. Export: `docs/PED/exports/PED_v3.0-review.pdf`. This is a review copy, with final SHA/signatures/independent approvals pending. Historical M1/M2 statuses above remain unchanged. Current M3 responsibilities: Don - architecture/core/configuration; Masego - verification/traceability; Emile - staging/performance/release/PED.

M3 document-control update - 17 October 2026

The preceding dated entries remain the historical record. PR #85 merged the supervised M3 implementation slice into `main` at `c7e92de8f818b66ca1ecc0faf88b0987c4b17b40`. Don and Masego approved PR #85 head `e76b96e74515e5cc813c6fba4cd1805362e5c020` before that merge; their approvals were not signatures on this later full-document export. PR #88 recorded DEC-013 through DEC-018, and PR #89 regenerated the earlier export. The decision-log gap is therefore closed; ADR-001/007 remain Proposed and DEC-008 remains Deferred. No M2 baseline promotion is inferred.

The earlier four-page PDF rendered section 22 alone. The corrected builder includes all existing numbered PED sources from 00 through 22, including M1/M2, a contents page and readable register records. Section 18 has no source and remains an explicit gap. This correction preserves historical signatures and baseline extracts. Current implementation/test evidence is linked from section 22 and the live controlled registers, not substituted into the historical M1 extracts.

PED version: **v3.0**. Team: **Emile, Don and Masego**. Repository: https://github.com/DonLukus/Civic-connect. Application implementation reference: the PR #85 merge above. RTM: `docs/requirements/RTM.csv`. Test/CI locations and simulated staging demonstration instructions: section 22 and `docs/deployment/m3-staging.md`. Record the actual submission date and final documentation commit/tag when submitting; this document does not claim a submission has occurred.

This full export and its wording corrections require their own normal PR review. Prior implementation approvals are not reused as approvals of new content. Public production release remains NOT READY; the reviewed M3 evaluation slice and public production are separate scopes.

M4 closure review update - 18 October 2026

Group U; official deadline 18 October 2026 (provided by Emile). Section 23 continues the same PED into a **v4.0 review candidate**, preserving historical M1/M2/M3 records. Current acceptance candidate: PR #103 revision `45bcdf6`, continuing verified application candidate `98b1c55700d940f915d856119f581adee8044986`. Original-AC QA calculation: 47.50% (19/40), subject to independent review; this is below the brief's 80% target. No original criterion was removed to improve the percentage. Final team deployment decision, assessed documentation reference and actual submission remain pending. No new signatures or M2 baseline promotion are inferred.

1. Introduction

Source: docs/PED/01-introduction.md

> **PED v1.0 section.** Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

This is the single evolving engineering record for CivicConnect [1, §6]. Version 1.0 establishes the baseline: what the team commits to build, for whom, within what constraints, what it has deliberately not decided, and how change will be controlled. It is extended, not replaced, at each milestone - v2.0 adds architecture and design, v3.0 construction and quality evidence, v4.0 the final success evaluation.

M1 is a foundation milestone. Technology stack, architecture, database schema, UI, API contracts, CI pipeline and deployment are outside its boundary. Section 14 records what is deferred and why.

1.1 Identifier scheme

Stable identifiers make traceability possible across four milestones (DEC-003). They are never reused or renumbered after baseline, because every later artefact attaches to them.

Prefix	Artefact	Prefix	Artefact
SH-nn	Stakeholder	RSK-nn	Risk
CF-nn	Stakeholder conflict	AS-nn	Assumption
SCOPE-I/O/D-nn	Scope: in, out, deferred	DEC-nnn	Engineering decision
CN-nn	Constraint	FEC-nn	Forward engineering consideration
FR-nnn / NFR-nnn	Requirement	AI-nnn	AI usage entry
AC-nnn / AC-Nnnn	Acceptance criterion	CHG / TST / DEF-nnn	Change, test, defect (reserved)
TR-nnn	Traceability record		

1.2 References

Ref	Source	Used for
[1]	SEN381 CivicConnect Master Project Brief v1.1 (2026)	Project-wide controls; scenario; minimum business capabilities
[2]	SEN381 Milestone 1 Brief (2026)	Required M1 outputs and the milestone gate
[3]	ISO/IEC/IEEE 29148:2018 - Requirements engineering	Requirement quality attributes applied in §6
[4]	ISO/IEC 25010:2011 - System and software quality models	Quality attribute categories structuring the NFRs
[5]	PMBOK Guide, 7th ed. (PMI, 2021)	Probability × impact risk scoring in §8
[6]	W3C WCAG 2.1 Level AA	Accessibility criteria in NFR-008

Citations support why a practice is appropriate; they are not evidence that a control was implemented. Implementation claims rest on repository history, register entries and configuration evidence [2, §9.1].

References (compiled §16 - full bibliography)

- Belgium Campus ITversity, 2026. *SEN381 CivicConnect Master Project Brief*, Version 1.1.

- Belgium Campus ITversity, 2026. *SEN381 Milestone 1 Brief: Engineering Foundation & Requirements Baseline*.

- International Organization for Standardization, 2018. *ISO/IEC/IEEE 29148:2018 Systems and software engineering - Life cycle processes - Requirements engineering*. Geneva: ISO.

- International Organization for Standardization, 2011. *ISO/IEC 25010:2011 Systems and software engineering - SQuaRE - System and software quality models*. Geneva: ISO.

- Project Management Institute, 2021. *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*, 7th ed. Newtown Square, PA: PMI.

- World Wide Web Consortium, 2018. *Web Content Accessibility Guidelines (WCAG) 2.1*. W3C Recommendation.

Before submitting: confirm this list matches the required referencing style and that every citation marker resolves to an entry. Missing in-text citations and references result in an automatic mark of zero [2, §10]. Verify each source independently and add any others the team actually consulted.

2. Problem Statement and Business Need

Source: docs/PED/02-problem-and-business-need.md

> **PED v1.0 section.** Owner: Masego. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

The organisation does not have a service request problem; it has an information control problem. Requests arrive by email, telephone, WhatsApp, spreadsheet and paper. Each channel holds a partial record, none holds the whole record, and none enforces a lifecycle. Every symptom follows from that single cause.

Symptom [1, §2]	Engineering cause	Addressed by
Requests duplicated, overlooked, misassigned or lost	No authoritative record; no controlled intake point	FR-005, FR-026
Requesters have limited visibility of status	Status held in people, not in a queryable system	FR-007, FR-008, FR-009
Staff cannot prioritise or identify ownership	No assignment model; no shared prioritised view	FR-011, FR-012, FR-014
Weak accountability for status changes	No audit of who changed what and when	FR-025, NFR-006
Management lacks outstanding/overdue information	No consistent state model, so aggregation is impossible	FR-019, FR-020, FR-021
Reporting is manual and hard to audit	Data re-keyed between channels; no single source	FR-020, FR-022
Sensitive information handled inconsistently	No access boundary around request content	FR-003, NFR-004, NFR-012
No controlled record of a request lifecycle	The lifecycle is undefined and therefore unenforceable	FR-015

This is why the scope baseline takes the shape it does. Consolidating onto one controlled record with an enforced lifecycle and an audit trail addresses the cause; adding channels would treat symptoms while worsening the cause, which is why SCOPE-O-02 excludes WhatsApp and SMS intake.

2.1 Business need and value

The organisation needs a controlled platform to submit, manage, monitor and report on service requests, improving visibility and accountability "without creating an unsustainable technical, operational or financial burden" [1, §2.1]. The second clause is a constraint, not an aspiration: it makes CN-03 binding, rules out paid always-on infrastructure, and forces the deferment in SCOPE-D-01.

Value	For	Judged by
Requests stop being lost - one record replaces five channels	SH-01, 02, 06	Every request has a reference and a state at all times (FR-005, FR-015)
Visibility without chasing	SH-01	A requester answers "where is my request" unaided (NFR-007)
Ownership becomes explicit	SH-02, 03	No request sits in a non-terminal state without an owner (FR-014)
Accountability becomes evidential	SH-04, 09	Any status change is attributable after the fact (FR-025, NFR-006)
Reporting stops being manual	SH-04	Dashboard figures reconcile to stored records (FR-019, FR-020)

Value	For	Judged by
Sensitive information is bounded	SH-09	Negative-access tests pass for every role (NFR-004, NFR-012)

2.2 Success beyond working code

Stated now so it can be evaluated honestly in M4 rather than invented then [1, §5]:

- Every request has a complete, attributable lifecycle record from submission to closure - evidenced by the audit trail, not asserted.

- A manager can answer what is outstanding, overdue and owned by whom from the system alone, and the figures reconcile to stored records.

- Delivered scope matches the approved baseline, and every deviation traces to an approved change request.

3. Stakeholder Analysis

Source: docs/PED/03-stakeholders.md

PED v1.0 section. Owner: Masego. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

ID	Stakeholder	Primary needs	Infl.	Int.
SH-01	Requester (community member / staff submitting)	Submit a request quickly; know it was received; see progress without phoning; be told when it is done	Low	High
SH-02	Service Staff / Technician	One prioritised queue; unambiguous ownership; full request detail; no duplicated or lost work	Medium	High
SH-03	Service Coordinator / Supervisor	Assign and rebalance work; see ageing requests; prevent unassigned backlog	High	High
SH-04	Operations Manager / Management oversight	Reliable and auditable information on outstanding, overdue, resolved and closed work; accountability for status changes; reporting that is not manual	High	High
SH-05	System Administrator	Manage users and roles; maintain the controlled category list; control security configuration	Medium	High
SH-06	Organisation / Project Sponsor	Improved visibility and accountability without unsustainable technical, operational or financial burden	High	Medium
SH-07	Development Team (SEN381 registered team)	Deliverable scope within schedule and actual team capability; defensible engineering evidence	High	High
SH-08	Lecturer / Assessor	Controlled, traceable and authentic engineering evidence produced progressively	High	High
SH-09	Data subject (requester whose sensitive information is held)	Confidentiality of security-related, personal or HR-sensitive request content	Low	High

Effort follows influence and interest. SH-03, SH-04, SH-07 and SH-08 are high on both and managed closely. SH-05 carries medium overall influence but high interest in its administrative domain - the controlled category list and role model that reporting (FR-020) and access control (NFR-004) depend on - and is managed closely there. SH-01 is low influence but high interest, and is the stakeholder most likely to abandon the system if visibility is poor - which is why status transparency is core scope rather than a convenience.

3.1 Competing expectations

A conflict is recorded only where the scenario evidences that both positions are genuinely held. Positions invented for balance were discarded (AI-006).

ID	Conflict	Resolution	Implemented by
CF-01	Requesters reporting security concerns want to submit anonymously to avoid reprisal. Management requires every request to be attributable for accountability and audit.	Authenticated submission only in v1. Requester identity is restricted to the assigned staff member, Manager and Administrator (NFR-012). An anonymous channel is explicitly deferred, not rejected.	DEC-006, FR-002, NFR-012, SCOPE-D-04, RSK-11

ID	Conflict	Resolution	Implemented by
CF-02	Staff want the smallest possible submission form so requests are actually logged. Management wants rich structured data so reporting is meaningful.	Four mandatory fields only (title, description, category, location). Category is mandatory and drawn from a controlled single-level list; every other structured field is optional. Reporting depth is bought with category discipline rather than form length.	FR-005, FR-006, FR-020, DEC-012
CF-03	Management expects SMS / WhatsApp escalation for overdue work. No paid messaging gateway is available within the cost constraint.	Notification limited to in-app and email in v1. SMS deferred. The unmet expectation is carried openly as a risk rather than silently dropped.	FR-009, SCOPE-D-03, RSK-08
CF-04	Requesters want to contact the assigned technician directly. Staff and privacy interests require that personal contact details are not exposed.	All communication occurs in an in-app comment thread attached to the request record. No personal contact details are surfaced to requesters.	FR-010, NFR-012
CF-05	The sponsor requires zero recurring cost. Free-tier hosting introduces cold-start latency and capacity ceilings that threaten the performance and availability expectations of daily users.	Cost constraint is treated as binding. NFR-001 is specified against a warm instance and cold-start behaviour is declared as a known limitation rather than hidden. The architecture decision that resolves this is deliberately deferred to M2.	CN-03, NFR-001, NFR-002, RSK-02, DEC-008

3.2 From need to requirement

Step	Content
Stakeholder	SH-04 Operations Manager (high influence, high interest)
Evidenced need	"Management has limited reliable information about outstanding, overdue and resolved work" [1, §2]
Requirement	FR-021 - identify a request as overdue when elapsed open time exceeds the configured target for its category
Acceptance criterion	AC-021 - open longer than target T in a non-terminal status is flagged; shorter is not. Tested at T−1, T, T+1
Dependency created	FR-023 must make the target configurable, or FR-021 has nothing to measure against
Open issue	Target values are not agreed with any stakeholder (RSK-05); automated escalation deferred (SCOPE-D-02)
Still testable because	FR-021 is specified against a configurable value, so boundary testing works now and only the values are provisional

One of forty such chains in the RTM. FR-025 is traced in full in Appendix A (see `docs/PED/07-traceability.md`).

4. Scope Baseline

Source: docs/PED/04-scope-baseline.md

> **PED v1.0 section.** Owner: Masego. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Baselined at v1.0; after baseline, scope changes only through a change request with impact analysis (§11.2). Excluded and deferred are separated deliberately - "we decided against this" and "we do not yet have evidence to decide" are different engineering statements.

4.1 In scope

ID	Capability committed	Requirements
SCOPE-I-01	Authentication and role-based access control for four roles: Requester, Staff, Manager, Administrator	FR-001..FR-004, NFR-003, NFR-004
SCOPE-I-02	Service request submission with a mandatory controlled category	FR-005, FR-006
SCOPE-I-03	Requester visibility: current status and full history of own requests	FR-007, FR-008
SCOPE-I-04	In-app and email notification on accept, reject, update and completion	FR-009
SCOPE-I-05	Staff work queue with search, filter and sort	FR-011, FR-012, FR-013
SCOPE-I-06	Assignment, self-assignment and reassignment of requests	FR-014
SCOPE-I-07	Controlled status lifecycle enforced as a state machine	FR-015, FR-017, FR-018
SCOPE-I-08	Action, comment and resolution recording on a request	FR-010, FR-016
SCOPE-I-09	Management dashboard covering open, in progress, overdue, resolved and closed work	FR-019, FR-021
SCOPE-I-10	Breakdown of activity by category, status, assignee and period, with CSV export	FR-020, FR-022
SCOPE-I-11	Administration of the controlled category list and of users and roles	FR-023, FR-024
SCOPE-I-12	Immutable audit trail of status and assignment changes	FR-025, NFR-006

4.2 Out of scope

ID	Excluded	Rationale
SCOPE-O-01	Native mobile applications (iOS / Android)	Responsive web delivery satisfies SH-01 access needs at a fraction of the build, test and release cost. Two additional deployment targets cannot be verified within the schedule.
SCOPE-O-02	WhatsApp, SMS or telephone channel integration	The fragmented-channel problem is solved by consolidating onto one controlled record, not by adding further channels. Paid gateways also breach CN-03.
SCOPE-O-03	Integration with existing finance, ERP, HR or asset-management systems	No interface contract, test environment or owning stakeholder is available. Integration risk is unbounded and unverifiable.
SCOPE-O-04	Payment, billing or procurement handling	No stakeholder need identified in the CivicConnect scenario.
SCOPE-O-05	Multi-language and localisation support	No stakeholder has stated a language need. Adds string management to every subsequent change.
SCOPE-O-06	Offline / disconnected operation with local synchronisation	Conflict resolution and sync integrity are disproportionate to the value at this scale.
SCOPE-O-07	AI-based automatic triage, classification or prioritisation of requests	Rejected as scope creep. No training data, no accuracy acceptance criterion, and it would place an unverifiable component on the critical accountability path.
SCOPE-O-08	External contractor / third-party supplier portal	Introduces an additional trust boundary and authorisation model that the security work in scope cannot cover.
SCOPE-O-09	IoT or sensor-originated automatic request creation	No such source exists in the described environment.

4.3 Deferred

ID	Item	Revisit	Reason and evidence required
SCOPE-D-01	File and photograph attachments on a request	M2 (pending DEC-007)	Requires object storage beyond the free tier, file-type and size validation, malware handling, and a retention/privacy obligation for images that may contain personal data. Deferred until the platform decision (DEC-008) establishes actual free-tier storage limits and until evidence exists of how often a photograph is genuinely required.
SCOPE-D-02	Automated SLA timers and escalation rules	M2	FR-021 currently identifies overdue work against a target resolution time per category. The target values have not been agreed with a stakeholder (RSK-05). Automating escalation on unvalidated thresholds would generate false escalations and erode trust in the system.
SCOPE-D-03	SMS / push notification channel	M4 or post-project	Returns when a funded messaging budget exists, or when a free-tier gateway is identified that meets the notification volume in NFR-013 at zero cost. Retained on the register so the unmet SH-04 escalation expectation (CF-03) stays visible; residual exposure is RSK-08.
SCOPE-D-04	Anonymous or confidential reporting channel	M2	Resolves CF-01 but requires a separate authorisation and audit model. Deferred pending a decision on whether accountability can be preserved without identity.
SCOPE-D-05	Bulk import of historical spreadsheet and paper records	M3	Data quality of the legacy records is unknown. Migration cannot be specified until the persistence design exists (M2) and a sample extract has been inspected.
SCOPE-D-06	Advanced analytics, trend reporting and scheduled report distribution	M4 / future	FR-019 to FR-022 satisfy the stated oversight need. Further analytics adds query and performance load against NFR-001 for value not yet evidenced.

4.4 A defended exclusion - no AI triage (SCOPE-O-07)

Automatic classification and prioritisation of requests was proposed during requirements drafting and rejected on four grounds:

1. No stakeholder need - it originated as an AI suggestion, not from the register (AI-002).

2. No acceptance criterion is possible. The team cannot state a defensible accuracy threshold or measure against labelled data, and a requirement without acceptance criteria cannot be baselined.

3. It would place an unverifiable component on the accountability path - the exact property the project exists to establish.

4. Inference at useful quality is not free, conflicting with CN-03, and creates a dependency the team does not control.

FR-026 was retained instead: a deterministic, testable duplicate warning on category and location. Same symptom, a fraction of the risk, and priced honestly as Could priority - the first requirement to drop under scope pressure.

5. Constraints and Assumptions

Source: docs/PED/05-constraints-assumptions.md

PED v1.0 section. Owner: Masego. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

ID	Type	Constraint	Engineering implication
CN-01	Scope	Committed scope must be baselined and controlled. Minimum business capabilities in the Master Project Brief section 3 must be delivered. Uncontrolled scope creep is not acceptable.	Every requirement must trace to a stakeholder source. Additional features enter only through a change request with impact analysis. Six items were deliberately deferred rather than absorbed.
CN-02	Schedule	Four assessed milestones inside a single SEN381 delivery period. Three part-time student engineers with concurrent modules.	M1 must be baselined before consequential architecture work begins, otherwise M2 inherits an unstable foundation. Work is split into three parallel workstreams with a fixed integration point rather than sequential handover.
CN-03	Cost / resources	Free or low-cost services only. Free-tier limits and likely operational cost beyond the educational context must be identified.	Rules out paid messaging gateways (CF-03), managed object storage at scale (SCOPE-D-01) and always-on compute. Free tiers idle out and cold-start, so: quality targets are set against a warm instance (NFR-001) with cold start declared as a known limitation (NFR-002); each separately deployed service is another instance that idles out, pushing M2 architecture towards a single deployable unit; M3 load testing must separate warm from cold measurements; and NFR-010 states operational cost at 100 users / 1 000 requests per month. Residual exposure is RSK-02. The chain is worked through in §5.1.
CN-04	Quality	Quality attributes must be defined now and supported by measurable evidence later. Unsupported claims are not accepted.	Every NFR carries a metric and a named verification method, even where the measurement itself only becomes possible in M3. NFRs without a measurement route were rewritten or removed.
CN-05	Security	Security is a lifecycle responsibility from requirements onwards, not a final hardening pass.	Authentication, authorisation, secrets handling, audit and privacy requirements are baselined in M1 so that M2 architecture cannot quietly design them out. Residual risk is recorded rather than a claim that the system is secure.
CN-06	Team capability	Three students of mixed and partly unknown proficiency. Learning-curve risk must be treated as an engineering constraint.	Technology selection is deferred to M2 so it can be made against an honest capability audit and a small proof of concept rather than familiarity or preference.
CN-07	Environment / platform	Belgium Campus cannot guarantee that any chosen language, framework, service or deployment platform is available or supported on the institutional desktop platform (Master Project Brief section 25).	Availability and compatibility must be verified on the actual build environment before the stack is committed in M2. A stack that only runs on one member's personal machine is a single point of failure.
CN-08	Governance	Protected main, pull requests for substantive change, and a minimum of two approvals from members other than the author. Self-approval is not accepted.	With a three-person team, two approvals means unanimous consent from the other two members. Any single absence blocks every merge, so a review turnaround rule and an escalation path are required from the start.

5.1 Constraint interaction - cost, quality, architecture

The interaction is where the engineering consequence lives, and this chain is why two decisions are deferred rather than taken.

Link	Effect
CN-03 binds	Only free-tier services. Free tiers idle out and cold-start, throttle compute and cap storage.
-> quality	NFR-001 cannot honestly promise 2 seconds on a cold instance, so it is specified against a warm instance at stated concurrency and volume; NFR-002 declares cold-start latency as a known limitation rather than concealing it.
-> architecture	Each independently deployed service is another instance that idles out, pushing M2 towards a single deployable unit. A financial constraint becomes an architectural one - to be argued in M2, not assumed now.
-> verification	M3 load testing must separate warm from cold measurements or the NFR-001 evidence is meaningless.
-> evaluation	NFR-010 requires operational cost at 100 users and 1 000 requests per month - where the free-tier assumption is honestly tested.
Residual risk	RSK-02 (16, High). Contingency is to publish the degraded behaviour or renegotiate CN-03 - not to quietly weaken the NFR.

A second interaction runs between CN-08 and CN-02: two approvals from non-authors means unanimity in a three-person team, so a single absence halts every controlled merge. The control cannot be weakened, so the schedule absorbs it - DEC-010 fixes a 24-hour review turnaround and RSK-04 carries the residual exposure.

5.2 Assumptions

ID	Assumption	Consequence if false	Risk	Validation
AS-01	The team acts as proxy for all CivicConnect stakeholders; no live client is available for validation.	Requirements may not reflect real operational practice	RSK-10	Validate with lecturer as proxy client at the M2 gate
AS-02	Provisional target resolution times per category may be set by the team and corrected later without breaching the baseline, because FR-021 is specified against a configurable value.	FR-021 acceptance testing is possible now; only the values are provisional	RSK-05	Confirm values at the M2 gate
AS-03	Email delivery is available at no cost within the free-tier limits of the platform selected in M2.	FR-009 becomes unimplementable as specified if untrue	RSK-02	Verify during the M2 platform spike
AS-04	Request volume during the academic project remains within the capacity stated in NFR-013.	Performance and cost evidence would need re-basing	RSK-02	Re-check at M3 load testing
AS-05	All three registered members remain on the team for the full delivery period.	Two-approval governance and workstream split both assume three members	RSK-04, RSK-09	Escalate to lecturer if it fails

6. Requirements and Acceptance Criteria

Source: docs/PED/06-requirements.md

> **PED v1.0 section.** Owner: Don. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Requirements follow the quality characteristics in ISO/IEC/IEEE 29148 [3]: necessary, unambiguous, singular, feasible, verifiable. Any requirement that could not be given a verification method was rewritten or removed - nine NFRs were rewritten because the drafts used unmeasurable language such as "fast", "user-friendly" and "secure" (AI-002, AI-003). Prioritisation is MoSCoW (DEC-009), with Must reserved for requirements without which the minimum business capabilities [1, §3] are not met; this makes the RSK-12 contingency executable, since scope is reduced in a defined order rather than renegotiated under pressure. NFRs are structured on ISO/IEC 25010 [4]: performance (001, 013), reliability (002, 011), security (003-006, 012, 014), usability (007, 008), maintainability (009), cost (010).

6.1 Functional requirements

FR-001	
ID	FR-001
Requirement	An Administrator shall create user accounts and assign exactly one role to each account; self-registered accounts shall default to the Requester role.
Source	SH-05
Pri.	Must
Acceptance criteria	Given an Administrator on the user administration screen, when a valid email and role are submitted, then an active account exists with that role and no other role, and the creation is written to the audit trail.
Verification	Functional test + audit record inspection

FR-002	
ID	FR-002
Requirement	The system shall authenticate a user by email address and password before granting access to any function other than sign-in and password reset.
Source	SH-05, SH-09
Pri.	Must
Acceptance criteria	Given an unauthenticated visitor, when any application URL other than sign-in or password reset is requested, then access is refused and the visitor is redirected to sign-in with no request data disclosed.
Verification	Automated negative-access test

FR-003	
ID	FR-003
Requirement	The system shall enforce four roles - Requester, Staff, Manager, Administrator - and permit each role only the operations authorised for it.
Source	SH-04, SH-09
Pri.	Must
Acceptance criteria	Given an authenticated user in role R, when an operation not authorised for R is attempted, then the operation is refused with an authorisation error and no state change occurs.
Verification	Automated role x operation matrix test (NFR-004)

FR-004	
ID	FR-004
Requirement	The system shall allow a user to reset a forgotten password through a single-use, time-limited link sent to the registered email address.
Source	SH-01
Pri.	Should
Acceptance criteria	Given a registered user requesting a reset, when the emailed link is used within its validity period, then a new password may be set; when the same link is reused or used after expiry, then it is refused.
Verification	Functional test, positive and negative

FR-005	
ID	FR-005
Requirement	A Requester shall submit a service request capturing title, description, category and location as mandatory information.
Source	SH-01
Pri.	Must
Acceptance criteria	Given an authenticated Requester with all four mandatory fields completed, when the request is submitted, then a request is stored with a unique reference, status New, the submitter recorded and a submission timestamp; and the reference is displayed to the Requester.
Verification	Functional test + database inspection

FR-006	
ID	FR-006
Requirement	The system shall require the Requester to select a category from a controlled list maintained by an Administrator, and shall reject free-text categories.
Source	SH-04
Pri.	Must
Acceptance criteria	Given the submission form, when no category is selected, then submission is refused with a field-level message; and the category selector offers only values from the active controlled list.
Verification	Functional test + boundary test

FR-007	
ID	FR-007
Requirement	A Requester shall view the current status of any request they submitted.
Source	SH-01
Pri.	Must
Acceptance criteria	Given a Requester with at least one submitted request, when the request is opened, then the current status, the date of the last status change and any resolution note are displayed.
Verification	Functional test

FR-008	
ID	FR-008
Requirement	A Requester shall view a list of all requests they have previously submitted, most recent first.

FR-008	
Source	SH-01
Pri.	Must
Acceptance criteria	Given a Requester with n submitted requests, when the request list is opened, then exactly those n requests are listed with reference, category, submission date and status, and no request belonging to another user appears.
Verification	Functional test + negative-access test

FR-009	
ID	FR-009
Requirement	The system shall notify the Requester in-app and by email when their request is accepted, rejected, updated with a comment, or completed.
Source	SH-01
Pri.	Must
Acceptance criteria	Given a request owned by Requester X, when its status changes to Accepted, Rejected or Resolved, or when a comment is added to the request, then an in-app notification and an email are generated to X within 5 minutes, containing the reference and either the new status and any reason, or the comment (CHG-001).
Verification	Functional test + notification log

FR-010	
ID	FR-010
Requirement	An authenticated Requester and authorised Staff shall exchange comments attached to a request record; personal contact details shall not be exposed.
Source	SH-01, SH-02
Pri.	Should
Acceptance criteria	Given a request with an assigned staff member, when either party adds a comment, then the comment is stored against the request with author and timestamp and is visible to both, and no email address or telephone number of the other party is displayed.
Verification	Functional test + privacy review (NFR-012)

FR-011	
ID	FR-011
Requirement	A Staff user shall view the service requests they are authorised to see, including unassigned requests and requests assigned to them.
Source	SH-02
Pri.	Must
Acceptance criteria	Given a Staff user, when the queue is opened, then unassigned requests and requests assigned to that user are listed, and requests assigned to another staff member are not shown by default.
Verification	Functional test

FR-012	
ID	FR-012
Requirement	A Staff or Manager user shall search by reference or keyword and filter and sort the request list by category, status, assignee, priority and submission date.

FR-012	
Source	SH-02
Pri.	Must
Acceptance criteria	Given a list of requests, when a filter or sort is applied, then only matching requests are returned in the requested order, and combining two filters returns the intersection.
Verification	Functional test + performance test (NFR-001)

FR-013	
ID	FR-013
Requirement	An authorised user shall view the complete detail of a request including its full status and assignment history.
Source	SH-02
Pri.	Must
Acceptance criteria	Given an authorised user opening a request, when the detail view is displayed, then submitted information, current status, assignee, all comments and the complete ordered change history are shown.
Verification	Functional test

FR-014	
ID	FR-014
Requirement	A Manager shall assign a request to a Staff user, a Staff user shall accept an unassigned request, and a Manager shall reassign an assigned request.
Source	SH-03
Pri.	Must
Acceptance criteria	Given an unassigned request, when a Staff user accepts it or a Manager assigns it, then that user becomes the assignee, the request status moves from New to Accepted and FR-009 is triggered, and the assignment is written to the audit trail; and when a Manager reassigns an already-assigned request, only the assignee changes, the status is unaffected, and the previous and new assignee are both recorded (CHG-001).
Verification	Functional test + audit record inspection

FR-015	
ID	FR-015
Requirement	The system shall permit only defined status transitions across New, Accepted, In Progress, On Hold, Resolved, Closed and Rejected, and shall refuse any undefined transition.
Source	SH-04
Pri.	Must
Acceptance criteria	Given a request in status Closed, when any transition other than the defined reopen path is attempted, then it is refused and the status is unchanged; and every permitted transition is recorded with actor and timestamp.
Verification	State-transition test covering the full permitted and forbidden matrix

FR-016	
ID	FR-016

FR-016	
Requirement	An assigned Staff user shall record actions taken and shall record resolution information when moving a request to Resolved.
Source	SH-02
Pri.	Must
Acceptance criteria	Given a request being moved to Resolved, when no resolution note is supplied, then the transition is refused; when a note is supplied, the note is stored against the request with author and timestamp.
Verification	Functional test, positive and negative

FR-017	
ID	FR-017
Requirement	An assigned Staff user shall move a request to Resolved, and a Manager shall move a Resolved request to Closed.
Source	SH-02, SH-04
Pri.	Must
Acceptance criteria	Given a Resolved request, when a Manager closes it, then the status becomes Closed with actor and timestamp recorded; and a Staff user attempting to close it directly is refused.
Verification	Functional test + authorisation test

FR-018	
ID	FR-018
Requirement	A Staff or Manager user shall reject or return a request to the Requester, and the system shall require a reason.
Source	SH-02
Pri.	Should
Acceptance criteria	Given a request being rejected, when no reason is supplied, then the rejection is refused; when a reason is supplied, the request status becomes Rejected, the reason is stored and the Requester is notified (FR-009).
Verification	Functional test, positive and negative

FR-019	
ID	FR-019
Requirement	A Manager shall view counts of open, in progress, overdue, resolved and closed requests for a selected period.
Source	SH-04
Pri.	Must
Acceptance criteria	Given a known data set for a period, when the dashboard is opened for that period, then each displayed count equals the count obtained directly from the stored records for that period.
Verification	Functional test against a seeded data set

FR-020	
ID	FR-020
Requirement	A Manager shall view request activity broken down by category, status, assignee and time period.
Source	SH-04

FR-020	
Pri.	Must
Acceptance criteria	Given a seeded data set, when a breakdown by category is requested, then every active category is represented and the sum of the category counts equals the total request count for the period.
Verification	Functional test with reconciliation check

FR-021	
ID	FR-021
Requirement	The system shall identify a request as overdue when its elapsed open time exceeds the configured target resolution time for its category.
Source	SH-04
Pri.	Must
Acceptance criteria	Given a category with a configured target of T hours and a request open for more than T hours in a non-terminal status, when the dashboard or queue is viewed, then that request is flagged overdue; and a request open for less than T hours is not flagged.
Verification	Boundary test at T-1, T and T+1. NOTE: target values are not yet agreed - see RSK-05 and SCOPE-D-02.

FR-022	
ID	FR-022
Requirement	A Manager shall export the filtered request list to CSV for offline analysis and record keeping.
Source	SH-04
Pri.	Should
Acceptance criteria	Given a filtered request list of n rows, when export is selected, then a CSV file is produced containing exactly n data rows with a header row and the same field values as displayed.
Verification	Functional test + row-count reconciliation

FR-023	
ID	FR-023
Requirement	An Administrator shall create, rename and deactivate categories, and shall configure the target resolution time for each category.
Source	SH-05
Pri.	Must
Acceptance criteria	Given a category referenced by existing requests, when deactivation is attempted, then the category is deactivated for new submissions while existing requests retain it; and deletion that would orphan a request is refused.
Verification	Functional test + referential-integrity test

FR-024	
ID	FR-024
Requirement	An Administrator shall change a user's role and deactivate a user account.
Source	SH-05
Pri.	Must

FR-024	
Acceptance criteria	Given an active user with assigned open requests, when the account is deactivated, then the user can no longer sign in, the assigned requests remain visible to the Manager for reassignment, and the change is written to the audit trail.
Verification	Functional test + audit record inspection

FR-025	
ID	FR-025
Requirement	The system shall record every status change and every assignment change with the acting user, timestamp, previous value and new value, and shall not permit these records to be edited or deleted through the application.
Source	SH-04, SH-09
Pri.	Must
Acceptance criteria	Given a request that has undergone k status or assignment changes, when the history is viewed, then exactly k ordered entries are shown each with actor, UTC timestamp, previous value and new value; and no application function exists that edits or deletes an entry.
Verification	Functional test + code review + negative test (NFR-006)

FR-026	
ID	FR-026
Requirement	On submission the system shall warn the Requester when an open request with the same category and location already exists, while still permitting submission.
Source	SH-02
Pri.	Could
Acceptance criteria	Given an existing open request with category C and location L, when a Requester submits a new request with category C and location L, then a warning listing the possible duplicate is displayed and the Requester may confirm or cancel.
Verification	Functional test

6.2 Non-functional requirements

NFR-001	
ID	NFR-001
Attribute	Performance - list response
Requirement	The 95th percentile server response time for a filtered request list of up to 500 records shall not exceed 2.0 seconds with 10 concurrent users on a warm instance.
Pri.	Must
Acceptance criteria	A load test of 10 concurrent users over 5 minutes against a 500-record data set reports a 95th percentile response time of 2.0 seconds or less. Cold-start latency is measured and reported separately (see NFR-002).
Verification	Load test, M3

NFR-002	
ID	NFR-002
Attribute	Availability

NFR-002	
Requirement	The staging deployment shall be available at least 95% of the time between 07:00 and 18:00 SAST on weekdays across the M3 evaluation period. Cold-start latency of up to 30 seconds after idle is a declared and accepted limitation of the free-tier hosting constraint.
Pri.	Should
Acceptance criteria	An uptime log across the evaluation period shows availability of 95% or more within the stated window, and the observed cold-start latency is recorded as a known limitation in the PED.
Verification	Uptime monitoring log, M3

NFR-003	
ID	NFR-003
Attribute	Credential storage
Requirement	Passwords shall be stored only as a salted adaptive hash (bcrypt with cost factor 10 or greater, or Argon2id). No password shall be stored in plaintext or in any reversible form.
Pri.	Must
Acceptance criteria	Inspection of the stored user records shows no plaintext or reversible password value, and code review confirms the hashing algorithm and cost factor.
Verification	Database inspection + code review, M3

NFR-004	
ID	NFR-004
Attribute	Authorisation enforcement
Requirement	100% of role-restricted operations shall reject a request from an unauthorised role with an authorisation error and without any state change.
Pri.	Must
Acceptance criteria	An automated test matrix covering every role against every restricted operation reports zero unauthorised successes.
Verification	Automated role x operation matrix test, M3

NFR-005	
ID	NFR-005
Attribute	Secrets handling
Requirement	No password, API key, token, connection string or private key shall be present in the repository or in its commit history.
Pri.	Must
Acceptance criteria	An automated secret scan on every pull request and a full-history scan both report zero findings.
Verification	Automated secret scan, from M1

NFR-006	
ID	NFR-006
Attribute	Auditability
Requirement	100% of status and assignment changes shall be recorded with actor, UTC timestamp, previous value and new value, and shall not be modifiable through the application.
Pri.	Must

NFR-006	
Acceptance criteria	For a scripted sequence of k changes, the audit record contains exactly k entries with all four fields populated, and no application endpoint accepts an update or delete against an audit entry.
Verification	Functional test + negative test, M3

NFR-007	
ID	NFR-007
Attribute	Usability of submission
Requirement	An untrained Requester shall complete a valid submission within 3 minutes and no more than 3 screens.
Pri.	Should
Acceptance criteria	In a moderated test with 5 representative participants, at least 4 complete a valid submission unaided within 3 minutes.
Verification	Moderated usability test, M3

NFR-008	
ID	NFR-008
Attribute	Accessibility
Requirement	Body text shall meet a contrast ratio of at least 4.5:1, every form input shall carry a programmatic label, and the submission flow shall be fully keyboard operable (WCAG 2.1 AA subset).
Pri.	Should
Acceptance criteria	An automated accessibility check reports zero contrast or missing-label violations on the submission and request-detail screens, and the submission flow is completed using keyboard input only.
Verification	Automated accessibility scan + manual keyboard walkthrough, M3

NFR-009	
ID	NFR-009
Attribute	Maintainability
Requirement	Automated unit tests shall cover at least 70% of lines in the business/service layer, and static analysis shall report zero critical findings on the main branch.
Pri.	Should
Acceptance criteria	The pipeline coverage report shows 70% or more line coverage for the business/service layer and the static-analysis stage reports zero critical findings.
Verification	CI report, M3

NFR-010	
ID	NFR-010
Attribute	Cost
Requirement	Recurring monetary cost shall be R0.00 for the duration of the academic project, and the projected operational cost at 100 active users and 1 000 requests per month shall be documented.
Pri.	Must
Acceptance criteria	A platform cost review lists every service in use, its free-tier limit, current consumption against that limit, and a costed projection at the stated volume.
Verification	Cost review, M2 and M4

NFR-011	
ID	NFR-011
Attribute	Recoverability
Requirement	A backup of the request data shall be taken at least daily, a documented restore procedure shall exist, and a restore shall be verified at least once before M4. Recovery point objective 24 hours; recovery time objective 4 hours.
Pri.	Should
Acceptance criteria	A backup schedule is evidenced, and a restore into a clean environment is performed and timed, producing a data set matching the source at the backup point within the stated RTO.
Verification	Documented restore rehearsal, M3/M4

NFR-012	
ID	NFR-012
Attribute	Privacy of requester information
Requirement	Requester contact details shall be visible only to the assigned Staff user, the Manager and the Administrator roles.
Pri.	Must
Acceptance criteria	Negative-access tests confirm that a Requester and an unassigned Staff user cannot retrieve another requester's contact details through any interface, including direct record access.
Verification	Automated negative-access test, M3

NFR-013	
ID	NFR-013
Attribute	Capacity
Requirement	The system shall hold at least 500 user accounts and 5 000 request records within the selected free-tier storage limits without breaching NFR-001.
Pri.	Should
Acceptance criteria	With 500 users and 5 000 requests seeded, storage consumption remains inside the documented free-tier limit and the NFR-001 load test still passes.
Verification	Capacity test, M3

NFR-014	
ID	NFR-014
Attribute	Input validation
Requirement	100% of user-supplied input shall be validated server-side; rejected input shall return a field-level message without discarding the data already entered.
Pri.	Must
Acceptance criteria	For each input field, a boundary and an invalid-type case are submitted with client-side validation bypassed; every case is rejected server-side with a field-level message and the retained form values are redisplayed.
Verification	Automated boundary and negative test, M3

6.3 Architecturally Significant Requirements (M2)

Nine requirements are tagged `ASR (M2)` directly on their row in `docs/requirements/functional-requirements.csv` and `non-functional-requirements.csv` (DEC-005 keeps the CSV as source of truth; this table is the narrative view). They're not a new identifier series - DEC-003 doesn't have one for ASRs, and these are existing FR-/NFR- rows, just flagged as ones that force a structural decision rather than being satisfiable by any reasonable implementation. Five actual drivers, not nine independent ones:

Driver	Requirements	Forces
Audit-trail immutability enforced against the application itself	FR-025, NFR-006	Transaction boundaries and persistence privileges - ADR-002
A controlled 7-state machine, cross-cut by role and field guards	FR-015	A dedicated status-lifecycle design decision instead of conditionals scattered wherever a transition can be triggered - ADR-003
Notification that must stay extensible to a channel not yet built (SCOPE-D-03)	FR-009	A notification mechanism not hard-wired to one channel - ADR-004, ADR-006
Performance and availability targets specified against a free-tier constraint (CN-03)	NFR-001, NFR-002, NFR-013	The architecture-style and deployment decision - a preference exercise becomes an engineering one once every extra service is a free-tier instance that idles out - ADR-001, ADR-005
Access control and privacy enforced against every interface, not the intended one	NFR-004, NFR-012	Where authorisation actually sits in the architecture, as a cross-cutting concern rather than a per-screen check - ADR-001

This is a starting set, not a ceiling - if the proof of concept or the architecture ADR itself surfaces another requirement that genuinely forces a structural choice, it gets tagged the same way, added here, and the reason stated.

Appendix B - Request Lifecycle States (compiled PED appendix)

The model underlying FR-015 - a requirements artefact stating which transitions are legitimate, not how they are implemented.

State	Meaning	May move to	Actor
New	Submitted, not triaged	Accepted, Rejected	Requester creates; Staff/Manager moves on
Accepted	Acknowledged as valid work	In Progress, On Hold, Rejected	Staff, Manager
In Progress	Assigned and being worked	On Hold, Resolved	Assigned Staff
On Hold	Blocked or awaiting information	In Progress, Rejected	Assigned Staff, Manager
Resolved	Work complete, resolution recorded	Closed, In Progress (reopen)	Staff resolves; Manager closes or reopens
Closed	Terminal - confirmed complete	None	Manager only
Rejected	Terminal - reason mandatory	None	Staff or Manager (FR-018)

Open question carried to M2 (FEC-02): whether Closed may be reopened and by whom. Left open deliberately - a business rule the team cannot authorise without a stakeholder, and guessing would embed an unvalidated rule in the acceptance criteria.

7. Requirements Traceability Matrix

Source: docs/PED/07-traceability.md

> **PED v1.0 section.** Owner: Don. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

All 40 baselined requirements are traced. Coverage is a control, not a statistic: a requirement absent from the matrix is one that nothing verifies, and a source producing no requirement is a need silently dropped. The four right-hand columns are deliberately empty - design references arrive in M2, issue, pull request and test identifiers in M3, release evidence in M4. Filling them now would be fabricated evidence.

TR-001	
Trace	TR-001
Source	SH-01 / Master Brief 2.0 - 'Requesters have limited visibility of whether a request was received'
Need	Requester must know a request was received and where it stands
Req.	FR-005
Pri.	Must
AC	AC-005
Verification	Functional test + DB inspection
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-002	
Trace	TR-002
Source	SH-01 / Master Brief 2.0
Need	Requester must know a request was received and where it stands
Req.	FR-007
Pri.	Must
AC	AC-007
Verification	Functional test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-003	
Trace	TR-003
Source	SH-01 / Master Brief 2.0
Need	Requester must know a request was received and where it stands

TR-003	
Req.	FR-008
Pri.	Must
AC	AC-008
Verification	Functional + negative-access test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-004	
Trace	TR-004
Source	SH-01 / Master Brief 3.0 requester capabilities
Need	Requester must receive meaningful feedback on acceptance, rejection, update and completion
Req.	FR-009
Pri.	Must
AC	AC-009
Verification	Functional test + notification log
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-005	
Trace	TR-005
Source	SH-04 / Master Brief 2.0 - 'Reporting is manual, inconsistent and difficult to audit'
Need	Management must identify outstanding and overdue work reliably
Req.	FR-019
Pri.	Must
AC	AC-019
Verification	Functional test on seeded data
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-006	
Trace	TR-006
Source	SH-04 / Master Brief 2.0
Need	Management must identify outstanding and overdue work reliably
Req.	FR-021
Pri.	Must
AC	AC-021
Verification	Boundary test at T-1, T, T+1
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined - threshold values provisional (RSK-05)

TR-007	
Trace	TR-007
Source	SH-04 / Master Brief 2.0 - 'weak accountability for changes to request status'
Need	Every status and assignment change must be attributable
Req.	FR-025
Pri.	Must
AC	AC-025
Verification	Functional + negative test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-008	
Trace	TR-008
Source	SH-04, SH-09 / CN-05
Need	Every status and assignment change must be attributable
Req.	NFR-006
Pri.	Must
AC	AC-N006
Verification	Functional + negative test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3

TR-008	
Release (M4)	Pending M4
Status	Baselined

TR-009	
Trace	TR-009
Source	SH-02 / Master Brief 2.0 - 'Staff have difficulty prioritising requests and identifying ownership'
Need	Staff must see one prioritised queue with clear ownership
Req.	FR-011
Pri.	Must
AC	AC-011
Verification	Functional test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-010	
Trace	TR-010
Source	SH-02 / Master Brief 3.0 staff capabilities
Need	Staff must locate relevant requests quickly
Req.	FR-012
Pri.	Must
AC	AC-012
Verification	Functional + performance test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-011	
Trace	TR-011
Source	SH-03 / Master Brief 3.0 staff capabilities
Need	Ownership of a request must be unambiguous and changeable
Req.	FR-014
Pri.	Must
AC	AC-014

TR-011	
Verification	Functional test + audit inspection
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-012	
Trace	TR-012
Source	SH-04 / Master Brief 2.0 - 'no single controlled record of the lifecycle'
Need	Status must move only through controlled transitions
Req.	FR-015
Pri.	Must
AC	AC-015
Verification	Full state-transition matrix test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-013	
Trace	TR-013
Source	SH-04 / CF-02
Need	Reporting must be possible by category without free-text noise
Req.	FR-006
Pri.	Must
AC	AC-006
Verification	Functional + boundary test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-014	
Trace	TR-014
Source	SH-04 / CF-02

TR-014	
Need	Reporting must be possible by category, status, assignee and period
Req.	FR-020
Pri.	Must
AC	AC-020
Verification	Functional test with reconciliation
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-015	
Trace	TR-015
Source	SH-09, CN-05 / CF-01
Need	Sensitive requester information must not be broadly visible
Req.	NFR-012
Pri.	Must
AC	AC-N012
Verification	Automated negative-access test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-016	
Trace	TR-016
Source	SH-05, CN-05
Need	Access must be restricted by role
Req.	FR-003
Pri.	Must
AC	AC-003
Verification	Role x operation matrix test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-017	
Trace	TR-017
Source	CN-05 / Master Brief 16
Need	Authorisation must be enforced, not assumed
Req.	NFR-004
Pri.	Must
AC	AC-N004
Verification	Automated matrix test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-018	
Trace	TR-018
Source	CN-05 / Master Brief 9
Need	No credential may exist in the repository
Req.	NFR-005
Pri.	Must
AC	AC-N005
Verification	Automated secret scan on every PR
Design (M2)	N/A - control active from M1
Issue/PR (M3)	Active from M1
Test (M3)	Secret scan run
Release (M4)	Pending M4
Status	Baselined - control operating

TR-019	
Trace	TR-019
Source	CN-03, SH-06 / Master Brief 4
Need	The solution must not create an unsustainable financial burden
Req.	NFR-010
Pri.	Must
AC	AC-N010
Verification	Cost review
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3

TR-019	
Release (M4)	Pending M4
Status	Baselined

TR-020	
Trace	TR-020
Source	SH-02, SH-04 / CN-04
Need	The system must remain responsive at realistic volume
Req.	NFR-001
Pri.	Must
AC	AC-N001
Verification	Load test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-021	
Trace	TR-021
Source	SH-05 / Master Brief 3.0 management capabilities
Need	Categories and their targets must be administrable
Req.	FR-023
Pri.	Must
AC	AC-023
Verification	Functional + referential-integrity test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-022	
Trace	TR-022
Source	SH-05
Need	Users and roles must be administrable
Req.	FR-024
Pri.	Must
AC	AC-024

TR-022	
Verification	Functional test + audit inspection
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-023	
Trace	TR-023
Source	SH-01, SH-02 / CF-04
Need	Requester and staff must communicate without exposing personal contact details
Req.	FR-010
Pri.	Should
AC	AC-010
Verification	Functional test + privacy review
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-024	
Trace	TR-024
Source	SH-02 / Master Brief 3.0 staff capabilities
Need	Actions and resolution must be recorded against the request
Req.	FR-016
Pri.	Must
AC	AC-016
Verification	Functional test, positive and negative
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-025	
Trace	TR-025
Source	SH-02 / Master Brief 2.0 - 'requests may be duplicated'

TR-025	
Need	Duplicate submissions should be reduced
Req.	FR-026
Pri.	Could
AC	AC-026
Verification	Functional test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined - lowest priority, first candidate for RSK-12 scope freeze

TR-026	
Trace	TR-026
Source	SH-05 / CN-05
Need	Accounts must exist and carry exactly one role
Req.	FR-001
Pri.	Must
AC	AC-001
Verification	Functional test + audit inspection
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-027	
Trace	TR-027
Source	SH-05, SH-09 / CN-05
Need	Only authenticated users may reach the system
Req.	FR-002
Pri.	Must
AC	AC-002
Verification	Automated negative-access test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-028	
Trace	TR-028
Source	SH-01
Need	A user locked out must be able to regain access safely
Req.	FR-004
Pri.	Should
AC	AC-004
Verification	Functional test, positive and negative
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-029	
Trace	TR-029
Source	SH-02 / Master Brief 3.0 staff capabilities
Need	Staff must see complete request detail including history
Req.	FR-013
Pri.	Must
AC	AC-013
Verification	Functional test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-030	
Trace	TR-030
Source	SH-02, SH-04 / Master Brief 3.0
Need	Requests must be closed only by an authorised role
Req.	FR-017
Pri.	Must
AC	AC-017
Verification	Functional + authorisation test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3

TR-030	
Release (M4)	Pending M4
Status	Baselined

TR-031	
Trace	TR-031
Source	SH-02 / Master Brief 3.0 requester capabilities
Need	A rejected request must carry a reason back to the requester
Req.	FR-018
Pri.	Should
AC	AC-018
Verification	Functional test, positive and negative
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-032	
Trace	TR-032
Source	SH-04 / Master Brief 2.0 - 'Reporting is manual'
Need	Management must be able to take activity data offline
Req.	FR-022
Pri.	Should
AC	AC-022
Verification	Functional + row-count reconciliation
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-033	
Trace	TR-033
Source	SH-01, SH-06 / CN-03
Need	The service must be reliably reachable during working hours
Req.	NFR-002
Pri.	Should
AC	AC-N002

TR-033	
Verification	Uptime monitoring log
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined - cold start declared as a known limitation (RSK-02)

TR-034	
Trace	TR-034
Source	SH-09 / CN-05
Need	Credentials must not be recoverable from storage
Req.	NFR-003
Pri.	Must
AC	AC-N003
Verification	Database inspection + code review
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-035	
Trace	TR-035
Source	SH-01 / Master Brief 3.0
Need	Submission must be achievable without training
Req.	NFR-007
Pri.	Should
AC	AC-N007
Verification	Moderated usability test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-036	
Trace	TR-036
Source	SH-01

TR-036	
Need	The service must be usable by people with visual or motor impairment
Req.	NFR-008
Pri.	Should
AC	AC-N008
Verification	Automated accessibility scan + keyboard walkthrough
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-037	
Trace	TR-037
Source	SH-07 / CN-04, CN-06
Need	The system must remain changeable by a three-person team
Req.	NFR-009
Pri.	Should
AC	AC-N009
Verification	CI coverage and static-analysis report
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-038	
Trace	TR-038
Source	SH-04, SH-06 / Master Brief 17
Need	Request data must survive loss of the environment
Req.	NFR-011
Pri.	Should
AC	AC-N011
Verification	Documented restore rehearsal
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-039	
Trace	TR-039
Source	SH-06 / CN-03
Need	The system must hold realistic volume inside free-tier limits
Req.	NFR-013
Pri.	Should
AC	AC-N013
Verification	Capacity test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

TR-040	
Trace	TR-040
Source	CN-05 / Master Brief 16
Need	Input must not be trusted from the client
Req.	NFR-014
Pri.	Must
AC	AC-N014
Verification	Automated boundary and negative test
Design (M2)	Pending M2
Issue/PR (M3)	Pending M3
Test (M3)	Pending M3
Release (M4)	Pending M4
Status	Baselined

Highlighted: TR-006 (FR-021) is baselined but carries a provisional threshold under RSK-05 - marked rather than presented as complete.

Appendix A - End-to-End Trace: FR-025 (compiled PED appendix)

FR-025 carries the accountability property that is the core business value, and shows how one stakeholder need propagates into requirements, quality attributes, design constraints, risks and future verification.

Stage	Evidence at v1.0
Source	[1, §2] "There is weak accountability for changes to request status and actions taken." SH-04, supported by SH-09.
Requirement	FR-025 - record every status and assignment change with actor, timestamp, previous and new value; do not permit editing or deletion through the application. Must.
Quality requirement	NFR-006 - 100% of such changes recorded with all four fields, not modifiable through the application.

Stage	Evidence at v1.0
Acceptance criteria	AC-025 and AC-N006: for k changes the history shows exactly k ordered entries with all four fields, and no application function edits or deletes one. The negative half matters most - this requirement is about what the system must refuse to do.
Verification	Functional test, negative test and code review (M3).
Trace records	TR-007 (FR-025), TR-008 (NFR-006).
Dependencies	FR-014 and FR-015 must emit change events; FR-013 exposes history to authorised users; NFR-012 bounds who may read it.
Constraint interaction	CN-03 - an append-only trail grows monotonically against free-tier row capacity, interacting with NFR-013. Retention is open (FEC-05).
Design constraint	FEC-02 - transition logic must be separable from the UI, or audit behaviour is only testable end to end and NFR-009 coverage becomes unaffordable.
Risk	RSK-11 - the trail concentrates sensitive history, so NFR-012 must bound read access.
Decision	DEC-006 - authenticated submission only; without an identity there is no actor to record, so CF-01 was resolved in a way that preserves this requirement.
Still to come	M2 persistence design and enforcement point; M3 issue, pull request, test identifiers and results; M4 release evidence.

A change to FR-025 after baseline would require revisiting NFR-006, FR-013, FR-014, FR-015, NFR-012, FEC-02, FEC-05, RSK-11 and DEC-006. The matrix makes that list discoverable rather than remembered.

8. Risk Register

Source: docs/PED/08-risk.md

> **PED v1.0 section.** Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Probability and impact rated 1-5 and multiplied [5]. Priority: Critical 20-25, High 12-19, Medium 6-11, Low 1-5. Mitigation reduces likelihood or impact beforehand; contingency is the response if it occurs anyway. Entries failing the specificity standard [1, §12] - "team communication problems", "technical difficulties" - were deleted rather than kept as padding (AI-004).

RSK-01	
ID	RSK-01
Risk	A formal client/lecturer change request arrives after the M1 baseline is approved and is absorbed informally, destroying baseline integrity
Cause	Milestone 3 explicitly mandates a lecturer change request and impact analysis, so this event is planned rather than merely possible
P	5
I	4
Sc	20
Priority	Critical
Mitigation	Change control procedure, template and CHG- identifier scheme established now, in M1, before the change arrives. Schedule contingency reserved in the M3 plan. Impact-analysis checklist covers requirements, design, data, security, test, schedule and risk.
Contingency	If the change cannot be absorbed, negotiate scope exchange against the deferred register (SCOPE-D-xx) rather than extend the schedule, and record the trade-off as a decision.
Owner	Emile
Links	CN-01, DEC-003

RSK-02	
ID	RSK-02
Risk	Free-tier hosting behaviour (idle shutdown, cold start, capacity ceilings) prevents the performance and availability NFRs from being met
Cause	Cost constraint CN-03 permits only free-tier services, which are not engineered for continuous availability
P	4
I	4
Sc	16
Priority	High
Mitigation	NFR-001 is specified against a warm instance and cold start is declared as a limitation rather than hidden. Free-tier characteristics are an explicit input to the M2 platform decision, with a measured spike required before commitment.
Contingency	Accept and publish degraded cold-start behaviour as a known limitation; or trade the constraint upward with the sponsor if a small paid tier becomes acceptable.
Owner	Don
Links	CN-03, NFR-001, NFR-002, DEC-008, FEC-01

RSK-03	
ID	RSK-03
Risk	The selected technology stack proves unavailable, unsupported or incompatible on the Belgium Campus desktop platform after work has begun
Cause	The Master Project Brief section 25 explicitly withdraws any guarantee of platform support
P	3
I	5
Sc	15
Priority	High
Mitigation	Technology decision deliberately deferred to M2 (DEC-008). Availability and compatibility must be verified on the actual institutional build environment, by more than one member, before the stack is committed.
Contingency	Fall back to a stack already proven on the institutional environment and record the rework cost as a decision consequence.
Owner	Don
Links	CN-07, DEC-008, FEC-01

RSK-04	
ID	RSK-04
Risk	The two-approval rule deadlocks the team: with three members, every substantive merge requires both other members, so a single absence halts all controlled work
Cause	Master Project Brief section 9 requires two approvals from members other than the author, and the team has exactly three members
P	4
I	3
Sc	12
Priority	High
Mitigation	Team Working Agreement fixes a 24-hour review turnaround, a daily review window, and small single-purpose pull requests so review is cheap. Work is partitioned so that no member is blocked on their own critical path by a single reviewer.
Contingency	Planned absence is announced at least 24 hours ahead and outstanding pull requests are reviewed before departure. Unplanned absence beyond 48 hours is escalated to the lecturer with evidence, rather than resolved by weakening the control.
Owner	Don
Links	CN-08, DEC-001, DEC-010

RSK-05	
ID	RSK-05
Risk	Target resolution times per category are not agreed, so FR-021 (overdue identification) has no validated threshold and its acceptance criteria cannot be finalised
Cause	No live client stakeholder is available to authorise service-level targets
P	4
I	3
Sc	12

RSK-05	
Priority	High
Mitigation	FR-021 is specified against a configurable per-category target (FR-023) rather than a hard-coded value, so the requirement remains testable at boundary values even while the values themselves are provisional. Automated escalation is deferred (SCOPE-D-02).
Contingency	Adopt provisional targets agreed with the lecturer as proxy client, record them as an assumption, and revisit at the M2 gate.
Owner	Masego
Links	FR-021, SCOPE-D-02, AS-02

RSK-06	
ID	RSK-06
Risk	A credential, connection string or API key is committed to the repository and remains recoverable in history
Cause	Multiple contributors, AI-assisted code generation that frequently embeds sample credentials, and configuration files added under time pressure
P	3
I	5
Sc	15
Priority	High
Mitigation	Secret scanning enforced on every pull request from M1. .gitignore covers environment and configuration files before any code exists. Configuration supplied through environment variables only. NFR-005 makes the control testable.
Contingency	Immediately revoke and rotate the exposed credential, then purge history. Treat the exposure as a realised risk with a decision record, not a silent fix.
Owner	Don
Links	CN-05, NFR-005, FEC-04

RSK-07	
ID	RSK-07
Risk	Unverified AI-generated content enters the controlled baseline and cannot be defended by the student who submitted it
Cause	All three members use AI assistants; AI output is fluent and therefore easy to accept without verification
P	4
I	4
Sc	16
Priority	High
Mitigation	AI Usage Register is mandatory for material contributions and records the verification actually performed. Pull request template requires the author to declare AI assistance. Reviewers check the register entry, not only the code. Every member must be able to explain any artefact they merged.
Contingency	Any artefact that its author cannot explain at review is reverted from the baseline and reworked, regardless of correctness.
Owner	Emile

RSK-07	
Links	CN-08, NFR-005, DEC-004

RSK-08	
ID	RSK-08
Risk	Stakeholder expectations that were deferred or excluded (SMS escalation, attachments, anonymity) resurface late as assumed scope
Cause	Deferred items are easily remembered as promised rather than as deferred
P	3
I	3
Sc	9
Priority	Medium
Mitigation	Deferred scope is held on a visible register with the reason and the evidence still required, rather than deleted. Every deferral names the milestone at which it is revisited.
Contingency	Route any resurfaced item through the change request process so its cost is made explicit before it is accepted.
Owner	Masego
Links	SCOPE-D-01..06, CF-03, CN-01

RSK-09	
ID	RSK-09
Risk	Contribution becomes uneven and one or more members cannot defend the shared artefacts individually
Cause	Workstream specialisation combined with time pressure encourages members to work only on their own section
P	4
I	4
Sc	16
Priority	High
Mitigation	Every member authors at least one pull request, reviews at least two, and owns issues in their own name. Cross-workstream review is compulsory, so each member reads the other two workstreams. A defence rehearsal is held before the milestone with questions directed across workstream boundaries.
Contingency	Reallocate work at the mid-point checkpoint and record the reallocation in the project history.
Owner	Masego
Links	CN-02, DEC-010

RSK-10	
ID	RSK-10
Risk	Requirements prove unstable because no real CivicConnect stakeholder exists to validate them
Cause	Stakeholders are represented by the team acting as proxy, informed only by the scenario description
P	3
I	3

RSK-10	
Sc	9
Priority	Medium
Mitigation	Every requirement records its source. Where a source is a team assumption rather than a stated stakeholder need, it is recorded on the assumption register and flagged for validation. Requirements are written to be testable independently of the proxy's opinion.
Contingency	Validate the highest-value assumptions with the lecturer as proxy client at the M2 gate and process any resulting corrections as change requests.
Owner	Don
Links	AS-01, RSK-05, CN-01

RSK-11	
ID	RSK-11
Risk	Sensitive request content (security concerns, personal circumstances) is disclosed to users who should not see it
Cause	Requests explicitly include security concerns and lost property; role boundaries are enforced in application code that has not yet been designed
P	3
I	5
Sc	15
Priority	High
Mitigation	NFR-012 and NFR-004 make the boundary testable before it is built. Privacy is a baselined requirement so the M2 design cannot omit it. Anonymity is deferred rather than improvised (SCOPE-D-04).
Contingency	Restrict the affected data element to the Administrator role, record the residual risk, and notify affected parties in line with organisational policy.
Owner	Don
Links	CF-01, NFR-012, NFR-004, FEC-04

RSK-12	
ID	RSK-12
Risk	Late M1 baseline compresses M2, so architecture decisions are made on an unstable foundation
Cause	Four milestones inside one delivery period leaves no float; M1 slippage propagates directly
P	3
I	4
Sc	12
Priority	High
Mitigation	M1 scope is deliberately bounded to foundation artefacts, with technology and architecture explicitly out of scope. Three parallel workstreams with one fixed integration point rather than sequential handover.
Contingency	If the baseline slips, freeze scope at the Must-priority requirements only and record the reduction as a decision with its consequence.
Owner	Emile
Links	CN-02, DEC-008

RSK-13	
ID	RSK-13
Risk	Critical knowledge (deployment, pipeline, database) is held by only one member
Cause	Workstream specialisation under schedule pressure
P	4
I	3
Sc	12
Priority	High
Mitigation	Cross-workstream review is compulsory. Every controlled procedure is written down in the repository rather than held informally. Each member must be able to trace one requirement end to end and explain one decision that is not their own.
Contingency	Pair the sole holder with a second member for one working session and record the transfer in the project history.
Owner	Masego
Links	CN-06, RSK-09, FEC-07

RSK-14	
ID	RSK-14
Risk	Concurrent editing of a single large binary PED produces merge conflicts and lost content, and makes review meaningless
Cause	Three authors editing one .docx; binary files cannot be merged or reviewed line by line
P	4
I	3
Sc	12
Priority	High
Mitigation	DEC-002: the PED is authored as modular Markdown in the repository, one file per section, and compiled to a single controlled document at baseline. Registers are held as CSV so that changes are reviewable as diffs.
Contingency	If a conflict occurs, the section owner reconstructs from the last approved version in history; the compiled document is never the source of truth.
Owner	Emile
Links	DEC-002, DEC-005, CN-08

8.1 The risk deserving most attention

RSK-01 - score 20, Critical - owner Emile. A change request arriving after baseline and being absorbed informally. Rated probability 5 because Milestone 3 explicitly mandates a lecturer change request: the event is planned, not merely possible. That rating makes the mitigation real. Because the change is certain, the change control procedure, impact analysis template and CHG- identifier scheme exist now - not designed under pressure when it arrives. The contingency is not "work harder". If the change cannot be absorbed, scope is exchanged against the deferred register and the trade-off recorded as a decision. Trading scope is controllable; silently compressing testing is not. RSK-02 (16) affects the product more, but is already managed by deferring the decision it threatens (DEC-008). RSK-01 threatens the control system itself, and every other artefact depends on the baseline holding.

9. Forward Engineering Considerations

Source: docs/PED/09-forward-engineering.md

> **PED v1.0 section.** Owner: Don. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Seven later lifecycle concerns that already influence current requirements, constraints, risks or assumptions [2, §4]. None is decided here - the register preserves later options rather than closing them.

FEC-01 - Deployment environment and free-tier operating characteristics

Why now	The cost constraint permits only free-tier hosting, whose idle-shutdown and capacity behaviour directly determines whether NFR-001 and NFR-002 are achievable. These NFRs are being baselined now, so the constraint shapes requirements before any deployment work begins.
Influences	M2 platform and architecture decision (DEC-008); M3 staging deployment; M4 operational cost review.
Missing information	Measured cold-start latency of candidate hosts; free-tier compute, storage and row limits; whether the platform is reachable and supported from the Belgium Campus desktop environment.
Risk of ignoring	Performance and availability targets are baselined that the eventual platform cannot meet, forcing either a requirements change after baseline or an undefended quality claim.
Links	RSK-02, RSK-03, CN-03, CN-07

FEC-02 - Testability of the request lifecycle

Why now	FR-015 defines a state machine and FR-025 an immutable audit trail. Both are only cheaply testable if the transition logic is separable from the user interface. That is an M2 design constraint which must be recorded now, while the requirements are being written, rather than discovered during M3 test design.
Influences	M2 component and layering decisions; M3 test strategy and coverage target in NFR-009.
Missing information	Which transitions are permitted from each terminal state, particularly whether Closed may be reopened and by whom; whether the audit trail is written by the application or enforced at the persistence layer.
Risk of ignoring	Transition logic embedded in controllers or pages forces slow end-to-end tests for behaviour that unit tests should cover, making NFR-009 unaffordable and regression testing unreliable.
Links	FR-015, FR-025, NFR-006, NFR-009

FEC-03 - Automated build, integration and quality gates

Why now	M3 requires automated build, test, static analysis and dependency checking. The quality gate defines what evidence must exist before a merge, so the definition of done and the acceptance criteria written now must be expressible as automated checks.
Influences	M2 technology choice, which determines available tooling; M3 pipeline implementation and quality-gate definition.
Missing information	Which build and test tooling the chosen stack supports; free-tier pipeline minute limits; whether a coverage gate can be enforced on the branch protection rule.
Risk of ignoring	Acceptance criteria are written that can only be verified manually, so quality evidence in M3 becomes screenshots and assertion rather than measurement.
Links	NFR-009, NFR-005, CN-04, DEC-008

FEC-04 - Security architecture, secrets and dependency exposure

Why now	Security requirements baselined in M1 (NFR-003 to NFR-006, NFR-012, NFR-014) constrain the M2 architecture. Secret handling and dependency vulnerability exposure begin the moment the first line of code exists, which is why the secret-scanning control is being enabled before construction rather than after.
Influences	M2 authentication, authorisation and trust-boundary design; M3 dependency scanning, security testing and configuration management.
Missing information	Where the trust boundaries actually sit once the architecture is known; which vulnerability-scanning tooling is available at zero cost; how configuration is injected per environment on the chosen platform.
Risk of ignoring	Security becomes a late hardening pass, which the Master Project Brief explicitly rejects, and a leaked credential in early history is expensive to purge and impossible to hide.
Links	RSK-06, RSK-11, NFR-005, CN-05

FEC-05 - Data model evolution, migration and recovery

Why now	The audit trail (FR-025) and the deferred legacy import (SCOPE-D-05) both constrain the persistence design. NFR-011 commits the team to a verified restore. Recovery cannot be retrofitted onto a schema that has no backup path within its free tier.
Influences	M2 persistence design; M3 migration approach and restore rehearsal; M4 operational readiness evidence.
Missing information	Free-tier backup and point-in-time-recovery capability of candidate databases; the quality and structure of the legacy spreadsheet and paper records; whether audit records are retained indefinitely or aged out under NFR-013 capacity limits.
Risk of ignoring	A schema is baselined that cannot support an append-only audit trail within free-tier row limits, or a restore that has never been rehearsed fails at the moment it is needed.
Links	NFR-011, NFR-013, FR-025, SCOPE-D-05

FEC-06 - Observability and operational readiness

Why now	M4 asks how the team would know the system is degrading before users complain. Logging and health checks must be designed in M2, not added in M4, and NFR-002 cannot be evidenced at all without an uptime measurement that has to be running throughout M3.
Influences	M2 logging and health-check design; M3 staging monitoring; M4 operational readiness and incident response evidence.
Missing information	What monitoring the chosen platform provides at zero cost; what log retention the free tier permits; what constitutes a meaningful health check for this application.
Risk of ignoring	NFR-002 has no evidence because nothing was measuring availability during the period it describes, and the M4 operational readiness claim is unsupported.
Links	NFR-002, NFR-011, CN-03

FEC-07 - Maintainability, technical debt and knowledge transfer

Why now	Only three engineers exist and each currently owns a distinct workstream. Debt incurred under M3 schedule pressure will be defended in M4, and a decision taken now to skip a control is a debt item whether or not it is recorded as one.
Influences	M2 design for changeability; M3 technical-debt register and rework decisions; M4 decision-consequence reflection.
Missing information	Which parts of the system are most likely to change for a new stakeholder need; what the team's actual review capacity is once construction begins alongside other modules.
Risk of ignoring	Debt accumulates unrecorded, so the M4 reflection becomes hindsight narrative rather than evidence, and single-point knowledge concentration (RSK-13) is discovered only when a member is unavailable.
Links	RSK-13, RSK-09, NFR-009, CN-06

10. Engineering Decision Log

Source: docs/PED/10-decision-log.md

> **PED v1.0 section.** Owner: Masego. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Ten decisions taken, two deliberately deferred. A decision not actually made is recorded as deferred with the evidence required, rather than presented as decided.

ID	Decision	Rationale	Trade-off accepted	Risk
DEC-001	A - single controlled repository, protected main, pull request required, two approvals from members other than the author, no self-approval, linear history.	It is the mandated control, it produces the review evidence the milestone assesses, and one repository keeps documentation and future code under a single traceable history.	With three members, two approvals means unanimous consent from the other two; this is slower and creates a deadlock exposure.	RSK-04
DEC-002	B - the PED is authored as numbered Markdown sections in docs/PED/ and compiled to PED_v1.0.docx for submission and sign-off.	Markdown produces line-level diffs, so a reviewer can see exactly what changed and approve a specific change. A binary .docx cannot be merged or meaningfully reviewed, and option C guarantees an integration crisis at the deadline.	An extra compile-and-format step is needed before each baseline, and the submitted document must be regenerated rather than edited directly.	RSK-14
DEC-003	A - FR-nnn, NFR-nnn, AC-nnn, SH-nn, CF-nn, CN-nn, RSK-nn, DEC-nnn, FEC-nn, AS-nn, CHG-nnn, TST-nnn, DEF-nnn. Identifiers are never reused and never renumbered.	Stable flat identifiers survive reprioritisation and deletion. Hierarchical identifiers break as soon as a requirement is split or moved, which is exactly what happens between M1 and M2.	Identifiers carry no meaning, so a reader must consult the register rather than infer relationships from the number.	Low
DEC-004	A - the AI Usage Register records material contributions with the verification actually performed, and the pull request template requires an AI declaration that the reviewer checks.	The assessed risk is unverified content entering the baseline, which applies at least as much to requirements and risk text as to code. Prohibition would be unenforceable and would waste a permitted engineering aid.	The register is only as good as the honesty of the entries, and maintaining it costs time on every substantive contribution.	RSK-07
DEC-005	A - registers are held as CSV under docs/, and a compiled .xlsx is generated for presentation and submission.	CSV changes appear as reviewable diffs, so a risk being downgraded or a requirement being edited is visible to reviewers. A workbook is opaque in review, and tables embedded only in the PED cannot be maintained between baselines.	Two representations must be kept in step, and the workbook must be regenerated rather than edited.	RSK-14
DEC-006	A - all submissions are authenticated in v1. Identity is protected by restricted visibility (NFR-012) rather than by absence of identity. A confidential channel is deferred (SCOPE-D-04).	Accountability and audit are the core business need in the CivicConnect scenario, and an anonymous request cannot be clarified, updated or fed back on, which breaks FR-009 and FR-010. Option C requires a second authorisation model that the M1 scope cannot support.	The reprisal concern raised by SH-01 is not fully resolved and remains open on the deferred register.	RSK-11
DEC-009	A - MoSCoW, with Must reserved for requirements without which the minimum business capabilities in Master Project Brief section 3 are not met.	MoSCoW makes the contingency in RSK-12 executable: under pressure the team drops Could and Should items in a defined order rather than negotiating individually. A numeric scale invites everything to be rated 5.	MoSCoW carries no ordering within a band, so sequencing still requires the backlog.	Low

ID	Decision	Rationale	Trade-off accepted	Risk
DEC-010	A - documented in docs/governance/team-working-agreement.md and agreed by all three members.	A fixed turnaround makes review latency predictable, and small pull requests make review cheap enough to actually perform properly. Batched review would concentrate risk immediately before the milestone.	Requires daily engagement from all three members regardless of whose workstream is active.	RSK-04, RSK-09
DEC-011	A - the Coordinator responsibilities of SH-03 are served by the Manager role in v1.	A fifth role multiplies the authorisation test matrix in NFR-004 for a separation of duty that no stakeholder has stated as a need. Scope discipline favours the smaller controlled model.	If assignment volume grows, Managers carry coordination workload that a dedicated role would absorb; SH-03 needs are met functionally but not organisationally.	Low
DEC-012	A - a single-level controlled list, administered under FR-023.	A single level keeps the submission form to one selector, keeps FR-020 reporting reconcilable, and keeps the target resolution time in FR-021 attached to exactly one level. Two levels double the administration and create ambiguous parent-level reporting.	Reporting granularity is limited to the top level, which SH-04 may find coarse once real volume exists.	Low

10.1 Deferred decisions

Deferring is acceptable and often correct where the team can state the evidence still required [2, §4]. Neither of these is an omission.

DEC-007 - File and photograph attachments

Context	Attachments would materially improve fault reporting, but introduce storage cost, malware exposure and a retention obligation.
Status	DEFERRED - no attachment upload or storage built in M2. Revisit at the start of M3, or sooner if the DEC-008 proof-of-concept completes first (reviewed 2026-09-30).
Why deferring is correct	Committing now would bind the M2 platform choice to a storage requirement whose cost is unknown, which inverts the correct decision order.
Consequence accepted	Fault reports remain text-only in the interim, which SH-02 may find limiting.
Evidence required	Free-tier object-storage limits of candidate platforms; frequency with which a photograph is genuinely needed; malware-scanning options at zero cost; retention rules for images containing personal data; and a confirmed requirement (FR ID and priority) for attachments, including size and type limits - none exists yet, since SCOPE-D-01 is deferred scope, not a baselined FR.
What it determines	Affects persistence design, upload validation, security testing and operational cost from M2 onwards. No attachment-related FR exists yet in the RTM; any file-upload security controls wait for the decision.
Risk	RSK-08, RSK-02

DEC-008 - Technology stack, architecture and deployment platform

Context	These are the most consequential technical decisions in the project and are explicitly outside the M1 boundary.
Status	DEFERRED to M2, to be taken through a weighted decision matrix and an Architecture Decision Record supported by a measured proof of concept.
Why deferring is correct	Evidence gathering has started, but the current PoC results do not compare the candidate stacks or explain the different persistence observations on Render and the BC desktop. Baselined requirements and NFRs are inputs to the decision, not outputs of it. The team still needs the remaining platform evidence and a reviewed comparison before selecting a stack.
Consequence accepted	The team cannot begin construction, which compresses M2. Accepted deliberately.
Evidence required	Gathered 2026-10-05: hosted Render login and request submission; test request #1 was unavailable at its confirmation route after two successful manual redeploys, including a repeat in Emile's signed-in Edge session (cause unknown). On BC Desktop, a repeat run on Python 3.14.5 installed `requirements.txt` successfully and passed all 15 tests in 5.88 seconds. `git rev-parse HEAD` and `git rev-parse origin/main` both returned `6beeb7a1353fdfdf8a9ed6966e1028b7fd3e0d75`. A separate local test request remained available after a Flask process restart using local SQLite; this does not explain the Render result. Still required: verify compatibility against the supported Belgium Campus environment; compare candidate-host free-tier limits and cold-start behaviour; complete the team capability audit; diagnose the Render persistence failure and decide how to verify persistence; review the combined evidence as a team. See `docs/decisions/poc-log.md`.
What it determines	Determines architecture, persistence, CI, deployment, operational cost and the achievability of NFR-001, NFR-002, NFR-010 and NFR-013.
Risk	RSK-02, RSK-03, RSK-12

10.2 M2 progress on DEC-008

The DEFERRED status above is untouched - this records evidence gathered so far, not a decision. The 2026-10-05 PoC log records the Render login/submission check and the BC Desktop test and process-restart check, including their limits. The Render request was unavailable after two successful redeploys; the second check in Emile's signed-in Edge session returned the same 404. The cause is not established. The BC result used local SQLite and does not explain the Render result. ADR-001 (`docs/decisions/adr/ADR-001-architecture-and-stack-options.md`) remains **Proposed, not Accepted**. The team must review the evidence and complete the remaining comparisons before deciding DEC-008 or accepting ADR-001.

10.3 Change requests since baseline

Formal changes to baselined requirements go through `docs/change/`, using the CHG- identifier series DEC-003 already reserves for exactly this. Listed here for discoverability; the change request itself is the authoritative record.

ID	Date	Requirements affected	Summary
CHG-001	2026-09-30	FR-009 (AC-009), FR-014 (AC-014)	Closes two acceptance-criteria gaps the team had already agreed on informally in Assignment 2 (§5.1) but never written back into the baseline: FR-009's AC now covers the comment-added notification trigger the requirement text already names; FR-014's AC now states explicitly that both a Staff self-accept and a Manager's initial assignment move a request from New to Accepted, while a Manager reassigning an already-assigned request only changes the assignee. See `docs/change/CHG-001-fr009-fr014-notification-and-acceptance-trigger.md`.

ID	Date	Requirements affected	Summary
CHG-002	2026-10-17	FR-001 (scope expansion)	Adds public Requester self-registration to the M3 delivery slice, beyond the original bootstrap-accounts-only plan. Formally agreed - Don's and Masego's decisions are each recorded as PR #86 comments, confirmed independently against the GitHub API, not just reported. FR-001 Administrator provisioning remains incomplete regardless. See `docs/change/CHG-002-m3-delivery-slice.md`.

10.4 M3 ADRs logged to the register

Six M2/M3 architecture and design decisions existed as ADRs for weeks before being formally logged here - a real gap, caught during M3 review (Issue #65). Logged 2026-10-17, reconciled directly against each ADR's actual current status rather than assumed:

ID	ADR	Status logged
DEC-013	ADR-001 (architecture style and stack)	Deferred - ADR-001 itself is still Proposed, not Accepted
DEC-014	ADR-002 (persistence and transaction boundary)	Decided - ADR-002 is Accepted
DEC-015	ADR-003 (lifecycle validation design pattern)	Decided - ADR-003 is Accepted
DEC-016	ADR-004 (notification fan-out design pattern)	Decided - ADR-004 is Accepted
DEC-017	ADR-006 (integration and worker placement)	Decided - ADR-006 is Accepted
DEC-018	ADR-007 (authentication/authorisation placement)	Deferred - ADR-007 itself is still Proposed, not Accepted

Logging DEC-013 and DEC-018 as Deferred records the register entry, not a status change - it does not move ADR-001 or ADR-007 to Accepted. That remains a separate team decision, undecided as of this entry.

11. Engineering Process and Team Governance

Source: docs/PED/11-process-governance.md

> **PED v1.0 section.** Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Short milestone-aligned increments with a controlled baseline at each gate. With three part-time engineers, the process elements that earn their cost are a shared issue backlog, small reviewable changes, a fixed review cadence and a formal gate.

11.1 Team working agreement (DEC-010)

Area	Agreement
Ownership	Masego: problem, stakeholders, scope, constraints, decisions. Don: requirements, acceptance criteria, RTM, forward engineering, repository setup and governance. Emile: risk, AI control, working agreement, document control, PED assembly and baseline. Ownership is drafting responsibility, not exclusive knowledge.
Review turnaround	24 hours, with a fixed daily review window so review is never the thing that waits.
Change size	Small and single-purpose. A change that cannot be reviewed properly in fifteen minutes is split.
Approval	Two approvals from members other than the author. No self-approval, no administrator bypass.
Absence	Planned absence announced 24 hours ahead with reviews cleared first. Unplanned absence beyond 48 hours is escalated to the lecturer rather than resolved by weakening the control (RSK-04).
Cross-review	Each member reviews both other workstreams - no member reviews only their own area, which would guarantee RSK-13.
Individual evidence	Each member authors >=1 pull request, meaningfully reviews >=2, owns issues, can trace one requirement end to end, and can explain one decision and one AI verification that are not their own.
Disagreement	Resolved by evidence. Where evidence is unavailable the decision is deferred with the evidence required stated - the standard applied to DEC-007 and DEC-008.
Done	Reviewed and approved by two members, identifiers consistent, RTM and affected registers updated, project history entry written.

11.2 Change control from this baseline

The procedure and template exist now, before the first change request arrives - the mitigation for RSK-01 [1, §14].

Step	Action	Evidence
1 Raise	Logged with a CHG- identifier, requester, date, expected value	Change request record
2 Impact	Requirements and acceptance criteria, architecture, UI, data and migration, API contracts, security and privacy, scope, schedule, cost, testing and regression, deployment and operations, risk and debt	Impact analysis
3 Decide	Accept, modify, defer or reject, with rationale	Decision log entry
4 Authorise	Two approvals from non-authors	Pull request approvals
5 Implement	On a branch referencing the CHG-identifier	Commits, pull request

Step	Action	Evidence
6 Verify	Acceptance criteria re-verified; regression considered	Test evidence
7 Baseline	PED version incremented; RTM and registers updated	New version, RTM diff

12. Repository and Configuration Management

Source: docs/PED/12-repository-configuration.md

> **PED v1.0 section.** Owner: Don. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

GitHub is an engineering control environment, not file storage. These controls have operated since the start of M1; the evidence is repository history rather than this description of it.

Control	Configuration	Evidence
Repository	One controlled team repository for documentation, registers and (from M3) source	Repository settings
Protected main	Direct pushes blocked; main is the controlled product state	Branch rule; a refused direct push
Pull requests	Required for every substantive change, including documentation	Merged pull request history
Approvals	Two, from members other than the author; self-approval refused	Approval records
Stale approvals	Dismissed on new commits, so approval refers to what merges	Branch rule
Conversation resolution	Required before merge	Resolved threads
Administrator bypass	Disabled - applies to every member including the owner	Branch rule
Force push / deletion	Blocked, so history cannot be rewritten	Branch rule
Secrets	Secret scanning and push protection on; .gitignore covers environment files before any code exists (NFR-005)	Scan results, .gitignore
Issues	Substantive work is an issue with an owner, linked to the closing pull request	Issue history
Conventions	Branches docs/, feat/FR-nnn-, fix/, chore/. Commits type(scope): summary (refs #issue)	Commit history

Two approvals matter because a substantive change alters the controlled state every later artefact depends on, and the author is least able to see what they missed - which matters most for AI-generated content, since it is fluent and therefore easy to accept unchecked. The cost is recorded honestly as RSK-04 rather than avoided.

13. Responsible AI Use

Source: docs/PED/13-responsible-ai.md

> **PED v1.0 section.** Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

All three members use AI assistants. AI output is not authoritative and does not transfer accountability [1, §10]. The control (DEC-004): material contributions are registered with the verification performed; the pull request template requires an AI declaration the reviewer checks; any member may be asked to explain any AI-assisted artefact, including one they did not author.

ID	Member	Task	Verification applied	Outcome
AI-001	Masego	Structuring the M1 artefact set against the Milestone 1 brief and Master Project Brief	Cross-checked every proposed artefact against Milestone 1 section 3 required outputs and the grading criteria. Two suggested artefacts with no assessed purpose were removed; the FEC register was restructured to carry the four fields the brief explicitly requires.	Accepted with modification
AI-002	Don	Drafting candidate functional and non-functional requirements from the CivicConnect scenario	Every requirement was traced back to a Master Project Brief section 3 capability or a section 2 stated problem. Requirements without a traceable source were deleted. Nine NFRs were rewritten because the generated versions used unmeasurable language such as 'fast', 'user-friendly' and 'secure'.	Accepted with substantial modification
AI-003	Don	Independent critique of the draft requirement set for testability	The flags were assessed by hand against the verification method column. Four acceptance criteria were rewritten to state a measurable outcome and a verification route; one flag was rejected because the criterion was already testable at boundary values.	Partially accepted
AI-004	Emile	Drafting the risk register and forward engineering considerations	Generic entries such as 'team communication problems' and 'technical difficulties' were deleted for failing the specificity standard in Master Project Brief section 12. Each retained risk was rewritten with a named cause and a contingency distinct from its mitigation. Probability and impact were re-rated by the team rather than accepted as generated.	Accepted with substantial modification
AI-005	Don	Drafting the GitHub governance procedure and pull request template	Each setting was applied in the live repository and verified by attempting a direct push to main and a self-approved merge; both were correctly refused. Screenshots retained alongside the settings evidence.	Accepted and verified in the live repository
AI-006	Masego	Analysis of stakeholder conflicts and scope exclusion rationale	Each proposed conflict was tested against the scenario text for evidence that both positions are genuinely held. One proposed conflict was discarded as invented. The resolutions were rewritten to name the artefact that implements them so the resolution is traceable rather than rhetorical.	Accepted with modification

> **AI-002 - a verified example.** A generated candidate requirement set was used as a starting point and was not accepted as written. Verified: every requirement traced back to a Master Project Brief §3 capability or §2 problem; anything untraceable was deleted. Changed: nine NFRs rewritten for using unmeasurable language, which fails the verifiability standard in [3]. Rejected: AI-based automatic triage - no stakeholder need, no definable acceptance criterion, and it would place an unverifiable component on the accountability path. Recorded as SCOPE-O-07 so the reasoning survives. Cross-checked: a second model (AI-003) found the original NFR-001 stated no concurrency and no data volume, making it untestable.

The register records what was rejected and changed, not only what was used. A register listing only successes evidences record-keeping, not verification.

14. M1 Boundaries

Source: docs/PED/14-m1-boundaries.md

> **PED v1.0 section.** Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Not decided	Why	When	Prepared
Technology stack	Requires compatibility verification, measured free-tier behaviour, a capability audit and a proof of concept - none exist yet (DEC-008)	M2	NFRs act as selection criteria; CN-06, CN-07; RSK-03 owned
Architecture	Follows from architecturally significant requirements, baselined here	M2	NFR-001, 002, 009, 013 identified as likely drivers
Database schema	Depends on the persistence decision; must hold an append-only audit trail within free-tier limits	M2	FEC-05; FR-025, NFR-006 constrain the design
UI design	Outside the M1 boundary; targets set first so design has criteria to meet	M2	NFR-007, NFR-008
API contracts	Depend on the architecture decision	M2	FEC-02 records the testability constraint
CI pipeline	Cannot be built before the stack is chosen	M3	FEC-03; NFR-005 secret scanning already active
Attachments	Storage cost, malware handling and retention unknown until the platform is chosen (DEC-007)	M2	SCOPE-D-01 with evidence required stated
SLA thresholds	No stakeholder authority exists to set service-level values (RSK-05)	M2	FR-021 configurable, so it stays testable

15. Baseline Sign-Off

Source: docs/PED/15-baseline-signoff.md

> **PED v1.0 section.** Owner: Emile. Authored here in Markdown and compiled to `docs/PED/exports/PED_v1.0.docx` (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

Field	Record
Project / baseline type	CivicConnect - engineering foundation and requirements baseline
Version and date	PED v1.0, 8 September 2026
Scope reviewed	YES - 12 in-scope areas, 9 exclusions, 6 deferments reviewed by all three; SCOPE-O-07 defended in §4.4
Requirements and traceability checked	YES - 40 requirements, all traced; each has source, priority and acceptance criteria; identifiers reconciled across all registers
Risk review completed	YES - 14 risks scored and owned (1 Critical, 11 High, 2 Medium); 5 assumptions linked to risks
Repository and governance checked	YES - protected main verified by a refused direct push and refused self-approval; two-approval rule evidenced on merged pull requests; secret scanning active
Outcome	ACCEPTED by the team as a sufficiently controlled foundation. The formal gate decision rests with the assessor.

15.1 Known limitations

Recorded deliberately - honesty about limitations is valued above unsupported claims of completeness [1, §23].

1. No live CivicConnect stakeholder exists; the team acts as proxy (AS-01, RSK-10). Requirements are testable, but their business validity is not independently confirmed.

2. Target resolution times underlying FR-021 are provisional (RSK-05). The requirement is testable at boundary values; the values are not authorised.

3. Every NFR states a metric and a verification method, but none has been measured. They are commitments, not evidence, until M3.

4. Task-level scheduling and effort estimation follow the M2 architecture decision.

15.2 Declaration

We confirm this document reflects the actual engineering work of the registered team; that AI assistance is recorded with the verification applied; that each of us reviewed the whole document, not only our own workstream; and that each of us can independently locate, explain and defend any artefact in this baseline.

M1 Baseline Sign-off

Member	Workstream	Signature	Date
Masego	A	[✓]	8 September 2026
Don	B	[✓]	8 September 2026
Emile	C	[✓]	8 September 2026

16. Architecture Baseline

Source: docs/PED/16-architecture-baseline.md

> **PED v2.0 section, new at M2.** Owner: Emile. Authored here in Markdown, compiled to `docs/PED/exports/PED_v2.0.docx` at the M2 baseline (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

16.1 M1 baseline review

Before adding any M2 decision, the whole M1 baseline (`docs/PED/01-15`, the decision log, the risk, requirements, scope and stakeholder registers) was read through end to end, as the M2 brief requires (§5.1). This is what that review found.

Materially changed:

- FR-009 and FR-014's acceptance criteria were clarified via CHG-001 - see `docs/PED/10-decision-log.md` §10.3. Both readings had already been agreed by the whole team in Assignment 2 but had never been written back into the baselined requirement text. Closed before the status-lifecycle and notification design decisions get built on top of the ambiguity.

Still open, not actioned here - needs input the team doesn't control:

- AS-01 and AS-02 both call for validating assumptions with the lecturer as proxy stakeholder "at the M2 gate" - target resolution-time values (FR-021, under RSK-05) and general requirement validity. That's a real conversation with the lecturer, not something resolvable by editing a document, so it stays open on the assumption register rather than being marked done.

Everything else: reviewed, unchanged, still holds. Identifiers are consistent across every register. DEC-001 through DEC-012 (except DEC-008, addressed below) still reflect the team's actual position - nothing in Assignment 2, Assignment 3 or the M2 brief surfaced a reason to revisit any of them. The scope baseline, risk register, stakeholder register and constraints are all still accurate as written. The Forward Engineering Considerations (`09-forward-engineering.md`) already name M2 as the milestone where FEC-01 through FEC-07 get acted on - that's expected, not a gap; this file and `17-20` are where that happens.

16.2 Architecture decision

ADR-001 remains Proposed and DEC-008 remains Deferred. The original request slice passed 10 tests on the Belgium Campus desktop on 30 September 2026. On 5 October, a fresh BC Desktop checkout passed 15 tests on Python 3.14.5; a later repeat installed `requirements.txt` successfully and again passed all 15 tests in 5.88 seconds on commit `6beeb7a1353fdfdf8a9ed6966e1028b7fd3e0d75` (matching `HEAD` and `origin/main`). A disposable local request remained available after a Flask process restart using local SQLite. The login/save slice was exercised on Render after PR #71. A test request was submitted, and its confirmation route returned 404 after two successful manual redeploys; the second check in Emile's signed-in Edge session returned the same result. The cause is unknown, and the BC local-file result does not explain the Render result. These observations are recorded in `docs/decisions/poc-log.md`; the repeated 404 narrows the evidence but does not explain the cause or complete the candidate-platform comparison, capability audit or persistence investigation needed for DEC-008. ADR-001 remains Proposed, not Accepted.

17. Data Persistence

Source: docs/PED/17-data-persistence.md

17.1 Purpose

This section captures the M2 persistence design for CivicConnect. It sets the authoritative persistence model for request data, audit data, and integration events, and it ties the design to the project requirements and quality constraints.

17.2 Data model

The core aggregate is the Request. A Request has one current state, one assignee, a version, and an immutable audit trail. The request aggregate is responsible for lifecycle legality and for tracking assignment changes.

Entities: - User - Request - Category - RequestComment - RequestAudit - OutboxEvent

Key fields: - Request.id - Request.version - Request.status - Request.assignee_id - Request.requester_id - Request.category_id - Request.created_at, updated_at - RequestAudit.request_id, actor_user_id, previous_status, new_status, previous_assignee_id, new_assignee_id, changed_at, correlation_id - OutboxEvent.correlation_id, aggregate_type, aggregate_id, event_type, payload_json, state, attempts, last_error, created_at

17.3 Correctness rule for FR-014 and FR-025

The real concurrency control is not a UI check. It is a conditional optimistic update with version verification.

Pseudo-SQL:

```
UPDATE requests
SET assignee_id = :staff_id,
    status = 'Accepted',
    version = version + 1,
    updated_at = CURRENT_TIMESTAMP
WHERE id = :request_id
  AND assignee_id IS NULL
  AND status = 'New'
  AND version = :expected_version;
```

If rowcount = 0, then another Staff member or transaction has already changed the request. The system rejects the second acceptance. This is the actual correctness mechanism protecting FR-014/FR-025.

17.4 Audit and outbox consistency design

The request-change transaction must be atomic across three writes: 1. update the Request row with version increment 2. insert the immutable RequestAudit row 3. insert the outbox row with the same correlation ID

This guarantees that the domain fact and the evidence of change stay together. The outbox row is the integration boundary and the audit row is the authoritative trace.

Outbox state machine: - pending -> sent -> failed - failed carries attempts and last_error - failed items may be retried without altering the request state

17.5 SPOF, backup and scalability notes

- There is a single primary database by default; the app tier remains stateless.

- Daily backup and periodic restore testing is required by NFR-011.

- Outbox processing is asynchronous and replayable so a notification outage does not break the business transaction.

- The design avoids a large synchronous fan-out on the request path and remains compatible with free-tier hosting constraints.

17.6 Decision summary

The persistence design is an adapted version of A2: relational versioned aggregate + immutable audit row + outbox. It preserves correctness, operational resilience, and project-fit for the current M2 constraints.

19. Design Decisions

Source: docs/PED/19-design-decisions.md

19.1 Lifecycle decision

The lifecycle problem is the real status-transition problem, not merely a UI state. FR-015..FR-018 constrain the legal movement of request status, and the same rules must be enforced in the domain service as well as in the UI.

Decision: use a transition-table validator rather than a pure State pattern.

Reason: - the rules are finite and explicit - the project needs easy review and tests - the legal transitions can be expressed in a table and enforced in a single validation step

19.2 Notification and audit fan-out decision

FR-009 and FR-025 require both requester communication and immutable auditability.

Decision: use observer-style fan-out, but do not roll back the status change when a notifier fails.

Reason: - the request state is the business fact - notification is a downstream delivery concern - email or in-app channel outages must not erase a valid lifecycle change

The failure handling is therefore: - change is committed - audit row is committed - outbox event is inserted as pending - notification is retried or marked failed

19.3 Persistence decision summary

The persistence layer uses a versioned aggregate plus immutable audit and outbox.

This is an adapted A2 solution with direct application to the current requirement set.

20. Integration and Deployment

Source: docs/PED/20-integration-deployment.md

20.1 Integration decision

Request state changes, audit records and notification delivery must integrate without losing correctness or auditability (FR-009, FR-025), under the free-tier cost constraint (CN-03). The full alternatives-considered analysis, rationale and consequences are recorded in **ADR-006** (`docs/decisions/adr/ADR-006-integration-decision.md`), not duplicated here per the M2 brief's instruction to reference decision evidence rather than restate it.

Summary: outbox-based asynchronous integration. The domain service writes the business change, its audit row and a pending outbox row in one local transaction; a separate process (currently `OutboxProcessor`, manually invoked - no scheduler yet, see the deployment direction below) publishes the event and marks it sent or failed. A downstream notification failure never rolls back the business fact - only the message-send attempt is retried.

20.2 Deployment direction

The current deployment direction - where the deployable unit runs, configuration and secrets handling, where state lives, and networking - is recorded in `docs/deployment/deployment-direction.md` and is **Proposed**, gated on the DEC-008 proof-of-concept, consistent with ADR-001. Deliberately deferred deployment decisions and the evidence still required are listed there rather than assumed here.

20.3 What is not yet established

- No scheduler for the outbox worker (§20.1) - invoked manually today.

- No verified hosting provider (§20.2) - SQLite vs. PostgreSQL for production is an open question, not a decision (see `docs/decisions/technology-versions.md` and RSK-15).

- PR #62 adds a session-gated login/save PoC, but this is a candidate branch rather than an accepted authentication/authorization design. ADR-007 and RSK-16 remain open for review; the branch has not been verified on the hosted service.

21. M2 Baseline Sign-Off

Source: docs/PED/21-m2-baseline-signoff.md

> **PED v2.0 section, new at M2.** Owner: Emile. Authored here in Markdown, compiled to `docs/PED/exports/PED_v2.0.docx` at the M2 baseline (DEC-002). The compiled document is an output - never edit it directly, and never treat it as the source of truth.

The full sign-off record - what is included, known limitations, open decisions and the three signatures - lives in `docs/baseline/M2-baseline-signoff.md`, not duplicated here, in the same way §20 references ADR-006 rather than restating it. This section narrates it for the PED's own continuity, as §15 did for the M1 sign-off.

21.1 Current outcome

As of the last recorded review (30 September - 1 October 2026), the baseline's own Outcome field reads **PROPOSED**, not ACCEPTED. All three members have signed the 30 September review; the 1 October evidence update (PoC log, PR #62, PR #63) is recorded but has not yet been separately re-approved by all three per that file's own note under "Outcome."

Two concrete reasons this stays PROPOSED, neither hidden:

1. ADR-001 (architecture and technology stack) and ADR-007 (authentication/authorization placement) are both still **Proposed**, not Accepted - the evidence DEC-008 requires (a login/save/deploy proof of concept verified on a live host, not assumed) is not yet complete. See `docs/decisions/poc-log.md`.

2. The decision register itself has not caught up with the ADRs it should be logging - see the gap recorded in `docs/PED/00-document-control.md`'s "Linked controlled artefacts" table and its graduation checklist.

21.2 What moves this to ACCEPTED

This section and `docs/PED/00-document-control.md` update together, in one pull request, once every condition in that document's "When v2.0 moves from Draft to BASELINED" checklist is true. Nothing here pre-empts that review - this section currently records PROPOSED because that is what the controlled sign-off file actually says, not because a different outcome is expected.

Dated finalisation clarification - 18 October 2026

The preceding Proposed outcome and signatures retain their historical scope. The decision-register reconciliation gap was addressed in PR #88 (DEC-013..018); that reason is no longer outstanding. ADR-001/007 remain Proposed and DEC-008 Deferred, so no new M2 acceptance or signature is inferred. Current product/scope consequences and the evidence-backed no-go recommendation are in PED section 23.

22. M3 construction, verification and release review

Source: docs/PED/22-m3-construction-verification-release.md

CivicConnect - PED v3.0, merged 16-17 October 2026. Prepared by Emile with AI assistance recorded in AI-036 and AI-038/042. Continues the existing M1/M2 PED sources; no historical signature or baseline status is changed.

22.1 Candidate and construction

Merged into `main` **at commit** `c7e92de8f818b66ca1ecc0faf88b0987c4b17b40` **via PR #85**, after Don and Masego approved PR head `e76b96e74515e5cc813c6fba4cd1805362e5c020`. The merge commit and the reviewed PR head are distinct identifiers. PR #85 folded together the backend routes, route tests, UI templates and dependency-check changes originally opened as PRs #80-84, plus PR #86 (the full role x operation authorisation matrix across all four roles, and the RTM scope-wording corrections both Don and Emile made after independent AI-assisted review). PRs #80, #82, #83 and #84 were closed as superseded once their content was verified present on `main` - not merged separately, to avoid duplicate/conflicting code. PR #81 (dependency-audit workflow) was independent of this chain and merged separately. Source revisions and file hashes are indexed in `docs/release/evidence/2026-10-16/source-manifest.json`. The registration addition is identified separately by `registration-source-manifest.json`.

Implemented slice: public Requester registration, login, submission, own list/detail/status; Staff login, queue/status filter/detail/acceptance; acceptance notification through in-app and SMTP; Manager/Administrator login with a safe landing page. `docs/deployment/m3-interface-agreement.md` records actual routes, forms, roles and error behaviour. `CHG-002` records the release-slice exclusions and impact; Don and Masego recorded acceptance for the supervised M3 evaluation. Baselined priorities and excluded obligations remain unchanged.

Two integration defects were corrected: missing CSRF tokens in shared logout forms on list/landing pages (DEF-004), and a mismatched audit-actor field in the detail template (DEF-005). Both have regression checks.

22.2 Architecture and configuration conformance

The candidate follows layered server-rendered Flask, service-level rules, SQLite persistence, transactional state/audit/outbox changes and an in-process notification worker. ADR-002/003/004/006 remain Accepted in the source; ADR-001/007 remain Proposed. DEC-008 remains Deferred. **The six Issue #65 decisions are now logged in** `docs/decisions/decision-log.csv` **as DEC-013 through DEC-018** (§10.4), reconciled directly against each ADR's actual current status - Deferred for DEC-013/018 (ADR-001/007, still Proposed), Decided for DEC-014/015/016/017 (ADR-002/003/004/006, Accepted). Logging them does not itself move ADR-001 or ADR-007 to Accepted; that is a separate, still-open team decision. Issue #65 is closed - the gap it raised (ADRs existing without a corresponding register entry) is fixed, not the underlying architecture/auth placement decisions themselves.

Staging is a distinct loopback Waitress process with a generated synthetic database, separate credentials and local STARTTLS SMTP sink. Config/secrets and the database stay Git-ignored. This simulates staging and does not establish public HTTPS, availability or production durability. See `docs/deployment/m3-staging.md`.

22.3 Verification results

Evidence	Result	Scope
pytest automated suite	60 passed (re-verified fresh in an isolated worktree, not just the commit message)	39 historical + 11 registration cases (registration-regression.xml) + 10 new role x operation matrix cases (PR #86)
Backend regression follow-up	16 passed	Includes corrected rendered audit actor assertion
Role x operation matrix	10/10 passed, including the previously-missing positive case (assigned Staff viewing their own accepted request)	`tests/test_m3_role_operation_matrix.py`; `docs/quality/role-operation-matrix.md`; closes the coverage gap Don's AI-assisted review found on the first integration commit
Ruff / Bandit / dependency audit	Clean / no medium-high findings / no known application dependency findings	Tool results at check time, not complete security proof
Live HTTP system journey 1	Passed	Requester login, submit #501, receipt, list/detail, logout and cross-requester denial
Live HTTP system journey 2	Passed	Staff filter/detail/accept, requester status, in-app notification and local SMTP capture
Browser walkthrough	Request #502 submitted and accepted	Actual browser UI operated; form/receipt screenshots retained; no checked-in browser automation claim
NFR-001 five-minute load	8,503 successful; 0 failures; p95 0.422s	10 distinct Staff users, 500 filtered records, local warm instance, target <=2s
Local process restart	Passed	Accepted #501 retrieved after stop/start of same-file staging
Online backup/separate restore	Passed	Integrity check OK; request/audit/outbox counts preserved

`docs/release/evidence/2026-10-16/` holds execution reports. The five-second smoke run is explicitly insufficient for NFR-001. The subsequent 300-second run supplies the stated measurement. The load/restart/backup evidence predates registration and was not rerun. Registration-live.json separately confirms public signup and login; email verification and signup rate limiting remain outstanding. No browser rendering benchmark or production-capacity claim is made.

22.4 Traceability and quality controls

`docs/requirements/RTM.csv` adds an execution-evidence column for the implemented slice without promoting baseline statuses. `docs/quality/test-inventory.md` remains the historical kickoff; `docs/quality/m3-verification-handoff.md` records the new boundary and remaining independent checks. Requirements, acceptance criteria, code, named tests, execution reports, DEF-004/005 and release conditions are connected through these artefacts.

GitHub workflows provide automated tests, static analysis, dependency audit and secret-pattern checks. Local results are not GitHub CI results. Current CI and branch-protection enforcement must be checked on the signed final revision. Two separate non-author students must review it; two accounts belonging to one person do not create two reviews.

22.5 Residual risk and release decision

Public production release: NOT READY. Don and Masego reviewed and approved the supervised M3 evaluation slice before PR #85 merged. Their CHG-002 decisions and the decision-register reconciliation are recorded; those completed steps are no longer pending. This corrected full PED export requires its own review and final documentation identifier. No new production-readiness classification, signature or M2 baseline acceptance is inferred. See the dated update in the release summary for the remaining conditions and owners.

Unresolved claims include durable hosted storage, external SMTP/TLS, dormant legacy-hash disposition, inactive-user/session edge cases beyond the documented role matrix, stuck/duplicate outbox recovery, reversible schema/code rollback, sustained availability, cold-start, capacity and accessibility. A local backup drill does not prove daily backup scheduling or retention. Local restart success does not close the M2 Render redeploy failure.

Render documents that free-service local files disappear on restart/redeploy/spindown, which is consistent with the observed M2 failure. The historical instance configuration was not independently recovered. Avoiding redeploy alone cannot ensure durability. PostgreSQL remains a candidate, not an adopted platform.

22.6 Evidence index and individual defence

Engineering concern	Authoritative artefact
M1 business/scope/requirements baseline	PED sections 01-15 and docs/requirements/ registers
M2 architecture/data/patterns/integration/baseline	PED sections 16, 17, 19, 20, 21 and docs/decisions/adr/
Current route/UI contract	docs/deployment/m3-interface-agreement.md
Controlled staging and recovery	docs/deployment/m3-staging.md; scripts/m3_staging.py; scripts/m3_verify_recovery.py
Test classifications and remaining matrix	docs/quality/m3-verification-handoff.md; tests/
Performance scenario and result	performance/load_test_staff_queue.py; performance.json
Live journeys and restart evidence	scripts/m3_verify_live.py; live-system-journeys.json; process-restart.json
Traceability and defects	docs/requirements/RTM.csv; docs/quality/defect-register.csv
Debt and residual risks	docs/quality/technical-debt-register.md; docs/release/m3-release-evidence-summary.md
Decisions/scope requiring review	docs/release/team-decision-review.md; docs/decisions/decision-log.csv (DEC-013..018); CHG-002
AI accountability and progressive history	docs/governance/ai-usage-register.csv; PROJECT_HISTORY.md; final PR/commit

Each student should be able to demonstrate: FR-014/AC-014 -> service version guard -> acceptance regression -> live journey #501 -> audit/outbox/notification evidence -> residual outbox recovery risk -> release condition. Emile should also explain NFR-001's dataset, concurrency, warm-up, duration, p95 method and local-only limits.

22.7 Final review actions - status as of 17 October 2026

1. ~~Emile inspects the candidate~~ - Reviews and test executions are recorded in PRs #85/#86, with AI assistance declared in the register.

2. ~~Don reviews backend, migration, outbox and decisions~~ - His AI-assisted reviews identified and corrected test-count, coverage, test-layer and scope-status claims. The operational limitations remain recorded.

3. ~~Masego records her CHG-002 decision/date~~ - Her acceptance is recorded under her account in PR #86, separately from Don's decision.

4. ~~Resolve the unconfigured Copilot review requirement~~ - The recorded resolution removed `copilot_code_review` from `main-protection-basic-02`; it did not rely on a merge bypass.

5. ~~Obtain two non-author approvals~~ - Don and Masego approved PR #85 head `e76b96e74515e5cc813c6fba4cd1805362e5c020`. The recorded seven CI checks passed.

6. ~~Merge the implementation~~ - PR #85 merged at `c7e92de8f818b66ca1ecc0faf88b0987c4b17b40`. PR #81 merged separately; PRs #80/82/83/84 closed as superseded after content checks.

7. ~~Log the six Issue #65 decisions~~ - DEC-013..018 appear in `decision-log.csv` and PED section 10.4; Issue #65 is closed.

Still genuinely open, not closed by any of the above: ADR-001 and ADR-007 remain Proposed - merging to `main` logs the decisions, it does not accept them. Durable hosted storage, external SMTP/TLS, dormant legacy password hashes, sustained availability, capacity and accessibility all remain unverified - see `docs/release/m3-release-evidence-summary.md`'s residual-risk table. This milestone's evaluation slice is complete and merged; a production release is a separate, later decision.

Sources checked 16 October 2026: https://flask.palletsprojects.com/en/stable/testing/ and https://render.com/docs/free. Repository artefacts above supply implementation and execution evidence.

23. Final scope, acceptance and project conclusion - PED v4.0 review candidate

Source: docs/PED/23-final-scope-and-project-conclusion.md

Group U. Deadline: 18 October 2026 (reported by Emile). Status: substantive closure prepared for independent review; not a signed submission or new team release decision. This section continues sections 00-22. Historical approvals retain their original scope.

23.1 Product and evidence identity

Active repository: https://github.com/DonLukus/Civic-connect. The current acceptance-quick-wins candidate is PR #103 revision `45bcdf6`, based on main `2524a66`; its final assessed commit/tag must be selected after independent review and merge. It continues the earlier verified application candidate `98b1c55700d940f915d856119f581adee8044986` with FR-006 validation, FR-022 filtered CSV export, FR-026 duplicate handling and DEF-007 mobile wrapping changes.

The two live HTTP journeys and local load/recovery reports are indexed in docs/release/evidence/2026-10-18/verification-summary.json. Results identify exact source hashes; no machine execution is attributed to a student personally. This is not a new feature programme.

Fresh verification on the assessed application candidate

Check	Actual observed result	Boundary
Automated suite	62 passed in 128.48 seconds	Includes 60 prior cases and two risk-driven final regressions; no browser-automation claim
CI suite and retained coverage	62 passed in 38.81 seconds; 78.50%, 70% threshold passed	Run 39091537311 on 52e175ba, Python 3.11; XML/JSON artifact 12122139293. Required-check settings remain separate.
Persistence line coverage	230/293 lines, 78.4983%	Includes lifecycle, outbox, earlier repository and both service modules
Five-minute warm list load	13,212/13,212 successes; zero failures; p95 0.281 seconds	10 Staff users, exactly 500 filtered records, 300 seconds, local Waitress; no cold-start/hosted-capacity claim
Two existing live HTTP journeys	Passed; notification observed in 1.125 seconds	Real loopback HTTP and local STARTTLS sink, not an external inbox or human browser session
Local stop/start	Accepted request 501 retained	Same isolated SQLite file; not cloud redeploy
Online backup/separate restore	Integrity OK; 501 requests, 502 audit rows, 1 outbox row preserved	Timed local drill; no daily schedule, retention or schema rollback claim
Static/dependency checks	Ruff clean; Bandit no findings; no known dependency findings	Tool/advisory scope at check time
Explicit history scan	Latest run: 179 commits scanned, no leaks found	Manual run 39091537311 on 52e175ba; earlier application run scanned 174. Existing exact false-positive ignore retained; separate from old zero-commit range run.
Accessibility/usability subset	Four submission fields labelled; four declared colour pairs >=4.5:1; Emile reported 5/5 moderated participants completed unaided within three minutes and no more than three screens	NFR-007 threshold met; no computed-style browser accessibility audit, so NFR-008 remains partial

23.2 Original baseline, controlled changes and scope result

The approved M1 record contains 26 FR, 14 NFR and 40 unique AC references. M1 is dated 8 September 2026. A pre-CHG-001 content snapshot at `52913f7121bfde410f1ec081665bd86a9ae80c98` was compared against current controlled CSVs: requirement wording and priorities are unchanged; only AC-009 and AC-014 differ, as documented by CHG-001. Snapshot provenance is not a fabricated historical release SHA.

CHG-002 approves a narrower supervised evaluation slice; it explicitly retains unfulfilled baseline obligations. No acceptance criterion is removed from this denominator. Original out-of-scope and deferred items are not added as failures. M2 remains Proposed where the baseline file says so; ADR-001/007 and DEC-008/013/018 do not become Accepted/Decided merely by merging code or this closure.

Proposed QA scope completion for the acceptance-quick-wins review candidate: 47.50% = 19/40 points. 12 criteria are assessed Met, 14 Partially Met and 14 Not Met. Met=1, Partially Met=0.5, Not Met=0. Scores are unweighted AC judgments, not test counts, and remain subject to independent review and merge. **The 80% target is not demonstrated.** The previous reviewed candidate was 40.00%; FR-022, FR-026 and the completed NFR-007 moderated result account for three additional points. The large original-scope variance is a material final outcome, not corrected by renaming partial delivery as complete. Every unmet Must criterion stays visible in the RTM.

23.3 M1 scope to final delivery

Original scope	Current outcome	Evidence and stakeholder consequence
SCOPE-I-01 - Authentication and role-based access control for four roles: Requester, Staff, Manager, Administrator	PARTIALLY IMPLEMENTED	Requester registration/login and four-role guards exist. Password reset and operational Administrator account functions remain absent.
SCOPE-I-02 - Service request submission with a mandatory controlled category	IMPLEMENTED for original submission ACs	Required fields, active controlled category, reference, New state and receipt have test/live evidence. Category administration remains a separate gap.
SCOPE-I-03 - Requester visibility: current status and full history of own requests	PARTIALLY IMPLEMENTED	Own list/detail/status exists; later resolution note/lifecycle commitments are incomplete.
SCOPE-I-04 - In-app and email notification on accept, reject, update and completion	PARTIALLY IMPLEMENTED	Acceptance in-app/local SMTP works; rejected/resolved/comment triggers and external inbox assurance remain incomplete.
SCOPE-I-05 - Staff work queue with search, filter and sort	PARTIALLY IMPLEMENTED	Staff queue/status filter/detail exists; broad search, combined filters, sorting and full comments/history are incomplete.
SCOPE-I-06 - Assignment, self-assignment and reassignment of requests	PARTIALLY IMPLEMENTED	Staff self-accept/version guard exists; Manager assignment and reassignment remain absent.
SCOPE-I-07 - Controlled status lifecycle enforced as a state machine	PARTIALLY IMPLEMENTED	Only New-to-Accepted is built. The complete seven-state permitted/forbidden matrix is not delivered.
SCOPE-I-08 - Action, comment and resolution recording on a request	NOT IMPLEMENTED	Actions, comments and resolution recording are unavailable; no full completion workflow.
SCOPE-I-09 - Management dashboard covering open, in progress, overdue, resolved and closed work	NOT IMPLEMENTED	Management open/overdue/resolved/closed counts and overdue identification are unavailable.
SCOPE-I-10 - Breakdown of activity by category, status, assignee and period, with CSV export	PARTIALLY IMPLEMENTED	A Manager can filter by status/category and export the exact displayed rows. Activity breakdown, period/assignee/priority filtering and broader reporting remain unavailable.
SCOPE-I-11 - Administration of the controlled category list and of users and roles	NOT IMPLEMENTED operational administration	Bootstrap seeds categories/users; it is not create/rename/deactivate/category-target or role/deactivation administration.
SCOPE-I-12 - Immutable audit trail of status and assignment changes	PARTIALLY IMPLEMENTED	Submission/acceptance audit exists with actor IDs; later transition/reassignment obligations remain incomplete.

Nine exclusions and six deferrals remain separately recorded in the existing scope baseline. All 14 NFRs are included in the AC denominator, even where they are not listed in a functional scope group. No untested NFR is silently excluded.

23.4 Criterion-by-criterion outcome

The existing RTM owns the detailed final QA fields; this table is its generated PED view, not a new RTM. Full original/current AC wording remains in section 06 and the controlled CSVs.

Requirement / AC	Priority	Proposed outcome / points	Evidence or limiting conclusion
FR-001 / AC-001	Must	NOT MET / 0	Original AC is Administrator account creation/one-role/audit: not implemented. Self-registration is delivered but is not that AC.
FR-002 / AC-002	Must	PARTIALLY MET / 0.5	Login/session checks work; public registration is an approved demo addition but the original sign-in/reset-only AC has not been formally rewritten. Partial pending explicit final AC reconciliation.
FR-003 / AC-003	Must	MET / 1	Four roles and existing-operation matrix verified; TST-052 also confirms deactivation revokes access. This does not supply missing operational features.
FR-004 / AC-004	Should	NOT MET / 0	Single-use, expiring emailed password reset is not implemented. Implement only if prioritised within remaining approved work; otherwise record unmet status and user consequence.
FR-005 / AC-005	Must	MET / 1	Required fields, controlled category, New state, submitter/reference/time and receipt have existing tests and live HTTP evidence.
FR-006 / AC-006	Must	MET / 1	Active-list selection and invalid/missing-category rejection verified; the field-level error is programmatically associated with the selector and other values are retained. Administrator maintenance remains a separate unmet FR-023 obligation.
FR-007 / AC-007	Must	PARTIALLY MET / 0.5	Own list/detail/status exists. Confirm status-change date and the original resolution-note clause; later resolution functionality remains outside the current slice.
FR-008 / AC-008	Must	MET / 1	Own recent-first list, fields and isolation are implemented and covered; stale TR-003 missing-list claim superseded.
FR-009 / AC-009	Must	PARTIALLY MET / 0.5	Acceptance in-app/local-SMTP evidence exists. Rejected, resolved and comment triggers plus intended external delivery remain incomplete/unverified; use CHG-001 AC wording.
FR-010 / AC-010	Should	NOT MET / 0	Requester/assigned-Staff comments with author/time and contact privacy are not implemented.
FR-011 / AC-011	Must	MET / 1	Unassigned and own-assigned Staff queue partitions have passing service/route evidence.
FR-012 / AC-012	Must	PARTIALLY MET / 0.5	Status filtering exists; reference/keyword search, combined category/assignee/priority/date filters and requested sorting remain incomplete.
FR-013 / AC-013	Must	PARTIALLY MET / 0.5	Detail and creation/acceptance history exist. Comments and complete later lifecycle/assignment history are not fully delivered.
FR-014 / AC-014	Must	PARTIALLY MET / 0.5	Staff self-accept/version check exists. Manager assignment and reassignment, unaffected status on reassignment and prior/new assignee audit remain incomplete.
FR-015 / AC-015	Must	PARTIALLY MET / 0.5	Only New-to-Accepted is implemented/specified for the slice. Remaining legal/forbidden transitions and original Closed/reopen AC are not established.
FR-016 / AC-016	Must	NOT MET / 0	Staff action/resolution recording and required resolution note are not implemented.
FR-017 / AC-017	Must	NOT MET / 0	Staff resolution and Manager closure with role negatives/audit are not implemented.
FR-018 / AC-018	Should	NOT MET / 0	Reject/return with mandatory reason and linked Requester notification is not implemented.
FR-019 / AC-019	Must	NOT MET / 0	Period-based management dashboard counts are not implemented.

Requirement / AC	Priority	Proposed outcome / points	Evidence or limiting conclusion
FR-020 / AC-020	Must	NOT MET / 0	Category/status/assignee/time-period activity breakdown and reconciliation are not implemented.
FR-021 / AC-021	Must	NOT MET / 0	Overdue identification against category targets is not implemented; target authority remains RSK-05. Do not invent authorised values.
FR-022 / AC-022	Should	MET / 1	Manager-only status/category filtered list and CSV export reuse the same six-field display mapping; TST-063 reconciles exactly n displayed rows to n CSV data rows plus one header. TST-064/065 cover role and invalid-input negatives. Broader FR-012 reporting/search remains partial.
FR-023 / AC-023	Must	NOT MET / 0	Operational category create/rename/deactivate/target management and referential-integrity behaviour are not implemented.
FR-024 / AC-024	Must	NOT MET / 0	Operational role change/deactivation, loss of sign-in, Manager reassignment visibility and change audit are not implemented.
FR-025 / AC-025	Must	PARTIALLY MET / 0.5	Creation/acceptance audit exists. Complete lifecycle/reassignment coverage remains partial; finalise k-change counting, actor/UTC/prior/new values and edit/delete negative evidence.
FR-026 / AC-026	Could	MET / 1	Same active category plus case-insensitive trimmed location against open statuses produces a warning listing safe request references, writes nothing before a choice, and permits confirm or cancel. TST-061/062 pass; review/merge remain pending.
NFR-001 / AC-N001	Must	PARTIALLY MET / 0.5	Five-minute 500-record/10-user warm local scenario is rerun/confirmed separately in the evidence summary. Cold-start measurement is still absent; full compound AC therefore partial.
NFR-002 / AC-N002	Should	NOT MET / 0	Weekday 07:00-18:00 SAST, >=95% staging availability over the M3 evaluation period and observed cold-start evidence are not established. State the real boundary; do not fabricate an elapsed-period uptime log.
NFR-003 / AC-N003	Must	PARTIALLY MET / 0.5	Fresh bcrypt accounts verified; legacy-hash fixture/login migration exists but dormant historical account storage has no accepted migration/reset disposition.
NFR-004 / AC-N004	Must	MET / 1	Existing-operation role matrix and new mid-session deactivation control pass. Missing operations are not counted as tested.
NFR-005 / AC-N005	Must	MET / 1	Manual quality run 39091537311 on 52e175ba scanned 179 commits with no leaks found, using the existing exact false-positive fingerprint ignore and no restricted log range. Earlier application-only run 39090851534 scanned 174 commits. Platform push protection/bypass and final review remain separate.
NFR-006 / AC-N006	Must	PARTIALLY MET / 0.5	Audit evidence covers the constructed slice. Confirm exact change sequence, actor/UTC/prior/new values and denied modification on final operations; later transitions/reassignments remain unavailable.
NFR-007 / AC-N007	Should	MET / 1	Emile supervised five different participants and reported 5/5 valid unaided completions within three minutes and no more than three screens: 2:12, 0:52, 0:37.29, 0:33.20 and 1:06.22; receipts #504-#508. This exceeds the required 4/5 threshold.
NFR-008 / AC-N008	Should	PARTIALLY MET / 0.5	Four labelled visible submission controls and four declared CSS contrast pairs pass a bounded automated check. Emile reported completing submission, duplicate confirmation and receipt by keyboard only with visible focus. A computed-style browser accessibility scan covering submission and detail still has not been run; partial only.

Requirement / AC	Priority	Proposed outcome / points	Evidence or limiting conclusion
NFR-009 / AC-N009	Should	MET / 1	The current candidate passes 96/96 local tests and 291/358 persistence lines (81.28%) against the 70% gate; Ruff is clean. All seven checks pass on PR #103 revision `45bcdf6`, including tests, coverage, Bandit, dependency audit, secret scans and traceability. Required-check settings and merge remain separate.
NFR-010 / AC-N010	Must	NOT MET / 0	Conclude R0 recurring-cost evidence, every service/free-tier limit/consumption and projection at 100 active users / 1,000 monthly requests. Resolve any storage/hosting-cost conflict without silently adopting a paid service.
NFR-011 / AC-N011	Should	PARTIALLY MET / 0.5	Fresh local online backup/separate restore passed in 0.031 seconds with integrity OK and matching 501 request, 502 audit and one outbox row counts. Daily scheduling/retention, hosted durability and recovery from an earlier point are not evidenced; the complete original AC remains partial.
NFR-012 / AC-N012	Must	MET / 1	DEF-006 nested email projection removed. Existing cross-requester/role checks plus TST-051 nested privacy and TST-052 inactive-session denial pass for the implemented interfaces.
NFR-013 / AC-N013	Should	NOT MET / 0	500 users plus 5,000 requests within documented storage limits while the NFR-001 test passes has not been demonstrated. Retain this distinction from the existing 500-request load result.
NFR-014 / AC-N014	Must	PARTIALLY MET / 0.5	Submission fields now have direct blank/tampered, non-text and over-limit server tests, field-associated messages and retained non-secret values (TST-058-060). Registration, CSRF and Manager filter negatives add evidence, but every implemented input still lacks both required boundary and invalid-type coverage.

23.5 M3 issue and register conclusion

DEF-004/005 fixes are merged and their automated regressions have been rerun. DEF-003's broad catch now logs exceptions but remains a retained error-classification concern; it is not falsely marked repaid. DEF-006 nested service data leakage is corrected by excluding unnecessary email projection; the current HTML template was not demonstrated to render that value. TST-051 failed before the fix and passes after it. TST-052 confirms deactivated users lose existing-session and direct-service access.

The existing risk and assumption CSVs now have dated closure-observation fields; they do not turn unverified mitigations into accepted facts. Existing TD-001..006 remain retained where recovery, durability, legacy hashes, missing functions or rollback still need work. The role-matrix and moderated-usability parts of TD-005 are addressed; computed-style accessibility, uptime and capacity limitations remain. The decision log retains actual Proposed/Deferred states and records consequences. AI-049/050/059 state actual assistance and checks, not new student votes.

23.6 Deployment / no-go assessment for team decision

QA recommendation: remain in isolated simulated staging; NO-GO for public production. Team decision: PENDING. The existing loopback Waitress/SQLite/local-STARTTLS environment is separated from development, uses synthetic accounts and ignored secrets and has reproducible commands. It is not public HTTPS, external mail or hosted persistence.

Blocking concern	Requirement/risk/debt	Consequence if released	What would change the decision
Hosted persistence not verified	NFR-011; TD-002; DEC-008	Request/audit data may be lost on host restart/redeploy	Select and verify durable storage and restore on the actual host
Outbox crash/retry recovery	FR-009; TD-001	Stuck notifications or duplicate external mail	Failure-injection evidence and an accepted recovery/idempotency policy

Blocking concern	Requirement/risk/debt	Consequence if released	What would change the decision
Historical credential and external transport boundaries	NFR-003; TD-003; FR-009	Inconsistent stored hashes or unverified external delivery/TLS	Inspect/migrate actual release data and verify intended SMTP environment
Broad baseline scope remains undelivered	Original Must requirements; CHG-002	Management, administration and lifecycle stakeholder needs remain unmet	Realistic completion or explicit accepted change/limitation; no invented completion score

This is a reasoned recommendation, not a new vote by Don or Masego. A justified no-go can satisfy the brief's deployment-decision expectation while still honestly reporting scope loss. No new paid host, database adoption, daily backup infrastructure or monitoring system is introduced solely for M4.

23.7 Significant decisions and project conclusion

1. A single server-rendered Flask deployment reduced UI/API integration overhead and enabled real HTTP journeys. It did not establish final hosting suitability; ADR-001 and DEC-008 remain open.

2. Transactional version/state/audit/outbox changes made acceptance race checks and notification failure separation demonstrable. In-process worker simplicity left stuck-sending/duplicate-delivery debt; an outbox is neither event sourcing nor exactly-once mail.

3. Restricting the supervised slice produced a coherent demonstration, but left many original Must commitments incomplete. The measured AC result must govern the final conclusion; reviewing a demo slice did not erase the original baseline.

The project delivered a useful, integrated Requester-to-Staff acceptance path with persistent local data, controlled access, attributable acceptance and measurable local evidence. It did not deliver the whole service-management product or prove public-production readiness. The most important next work is the unmet original workflow/management/admin scope, decision closure and durable/operational assurance, rather than optional new features. No assessment mark is predicted.

23.8 Combined demonstration and individual defence preparation

Use the final reviewed reference, not separate student prototypes. The brief's guide totals about 24 minutes: product journeys (6), M1-to-delivery scope and percentage (6), M3 fixes/limitations (4), selected acceptance traces (4), conclusion/decision consequences (4). Confirm the actual session allocation. Existing staging runbook supplies commands and locally generated synthetic credentials; do not put those credentials in the PED or slides.

Student	Evidence to navigate and explain
Emile	FR-014/AC-014 -> version guard -> acceptance test/live journey -> audit/outbox -> partial scope; NFR-001 workload and p95 method; continuing PDF and actual AI verification boundaries
Don	State/audit/outbox transaction, proposed vs accepted ADRs, host/durability limits, actual rulesets/CI and a meaningful failure/recovery trade-off
Masego	Original AC versus test assertions, role/ownership matrix and new regressions, final RTM/partial scores, defect closure and privacy/keyboard/usability evidence limits

Each student should show their actual contribution, one requirement-to-final-evidence trace, a decision outside their own component, one resolved and one retained issue, a failure/change response and a concise evidence-based conclusion. These are preparation notes, not a claim that rehearsal or human defence has occurred.

23.9 Final submission information and remaining gates

Field	Current record
Team name / number	Group U (provided by Emile)
Members	Emile, Don, Masego; confirm institutional identifiers if Moodle requires them
Repository	https://github.com/DonLukus/Civic-connect
Assessed application candidate	Review branch `m4/acceptance-quick-wins`, based on merged main `2524a66`; exact final commit remains pending review/merge
Final commit / tag / release	PENDING final review and team freeze; do not submit the candidate label as an approved release
PED	v4.0 review candidate, continuing the existing PED; promote to Final only after actual review
Final RTM	docs/requirements/RTM.csv - 40 original records with proposed final QA fields
Product access	Local staging demonstration per docs/deployment/m3-staging.md; no new public production URL
Official deadline	18 October 2026, supplied by Emile; exact Moodle cutoff time not supplied
Actual submission date / receipt	PENDING actual submission; no submission claimed

Before submission: obtain independent review, confirm/adjust AC judgments, record the real team deployment decision, freeze the exact assessed documentation/product reference, inspect the final PDF and check assessor access. Upload PED v4.0 Final, correct repository URL, exact final reference and any specifically required access information to Moodle; preserve the receipt. No separate code ZIP unless instructed. Do not change the assessed baseline after the deadline without explicit authorisation.